

MOTIVATE mapping - Data on extent and functional trends from RF

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Session info 89

This notebook analyses data on habitat extent metrics and functional trends derived from Random Forest models.

Load libraries

```
library(dplyr)
```

```
##
## Adjuntando el paquete: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(readr)
library(here)
```

```
## here() starts at C:/Users/jimenezalfaro/OneDrive - Universidad de Oviedo/IMIB/Analyses/MOTIVATE/MOTIV
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(ggplot2)
library(gghalves)
library(car)
```

```
## Cargando paquete requerido: carData
```

```
##
## Adjuntando el paquete: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##   recode
```

```
library(emmeans)
```

```
## Welcome to emmeans.  
## Caution: You lose important information if you filter this package's results.  
## See '? untidy'
```

```
library(multcomp)
```

```
## Cargando paquete requerido: mvtnorm  
  
## Cargando paquete requerido: survival  
  
## Cargando paquete requerido: TH.data  
  
## Cargando paquete requerido: MASS  
  
##  
## Adjuntando el paquete: 'MASS'  
  
## The following object is masked from 'package:dplyr':  
##  
##      select  
  
##  
## Adjuntando el paquete: 'TH.data'  
  
## The following object is masked from 'package:MASS':  
##  
##      geyser
```

```
library(ggeffects)  
library(patchwork)
```

```
##  
## Adjuntando el paquete: 'patchwork'  
  
## The following object is masked from 'package:MASS':  
##  
##      area
```

```
library(stringr)
```

Load data

```

# Data of plots with constant habitat
RF_data_nochange <- read_csv(here("data", "RF", "MOTIVATE_ALL_V1_SPECTRAL.csv"))

## Rows: 11519 Columns: 43
## -- Column specification -----
## Delimiter: ","
## chr (9): system_index, eunis1, biogeo, plot_id, geo, run_id, time_quartile,...
## dbl (33): area_first_total_ha, area_last_total_ha, delta_q_ha, delta_r_ha, d...
## lgl (1): eunis_cambio
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# Data of plots with habitat change
RF_data_change <- read_csv(here("data", "RF", "MOTIVATE_CHANGE_ALL_V1_SPECTRAL.csv"))

## Rows: 265 Columns: 47
## -- Column specification -----
## Delimiter: ","
## chr (11): system_index, eunis_first, eunis_last, biogeo, plot_id, transition...
## dbl (35): area_first_total_ha, area_last_total_ha, delta_q_ha, delta_r_ha, d...
## lgl (1): eunis_cambio
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

Load initial dataset from validation

```

initial_dataset <- read_csv(here("../", "MOTIVATE_validation", "data", "clean",
                                "ReSurvey_QRST_data_V2025.2_20261007.csv"))

## Rows: 120022 Columns: 58
## -- Column specification -----
## Delimiter: ","
## chr (40): Country, ReSurvey project, ReSurvey site, ReSurvey plot, ReSurvey...
## dbl (13): PlotObservationID, Lon_updated, Lat_updated, year, n_ESY_EUNIS, P...
## lgl (4): ESY_EUNISa_4, ESY_EUNISa_4_descr, multi_coords, n_to_n
## date (1): date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

Remove plots with habitat change from dataset of plots with constant habitat

```
RF_data_nochange <- RF_data_nochange %>%
  filter(!plot_id %in% RF_data_change$plot_id)
```

HERE: Check some columns that seem to be duplicated

```
RF_data_nochange %>% filter(coalesce(eunis1, "NA") != coalesce(EUNIS1, "NA"))
```

```
## # A tibble: 517 x 43
##   system_index area_first_total_ha area_last_total_ha delta_q_ha delta_r_ha
##   <chr>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 0000000000000000~ 77.7          77.7          2.81          5.28
## 2 0000000000000000~ 77.8          77.8         -4.06          5.68
## 3 0000000000000000~ 77.8          77.8        -27.0          29.6
## 4 0000000000000000~ 31.9          31.9          1.31         -0.126
## 5 0000000000000000~ 32.6          32.6          1.28        -0.0722
## 6 0000000000000000~ 77.9          77.9          6.72         -4.79
## 7 0000000000000000~ 77.9          77.9          5.16         -2.47
## 8 0000000000000000~ 77.8          77.8         -1.96         -2.58
## 9 0000000000000000~ 77.8          77.8        -29.4          26.2
## 10 0000000000000000~ 77.8          77.8          2.37          1.50
## # i 507 more rows
## # i 38 more variables: delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>,
## #   lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## #   r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## #   t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## #   last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, geo <chr>,
## #   run_id <chr>, years_between.x <dbl>, delta_focal_ha <dbl>, ...
```

```
RF_data_nochange %>% filter(is.na(eunis1))
```

```
## # A tibble: 0 x 43
## # i 43 variables: system_index <chr>, area_first_total_ha <dbl>,
## #   area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## #   delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## #   lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## #   r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## #   t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year.x <dbl>,
## #   n_resurvey <dbl>, plot_id <chr>, geo <chr>, run_id <chr>, ...
```

```
RF_data_nochange %>% filter(coalesce(biogeo, "NA") != coalesce(BIOGEO, "NA"))
```

```
## # A tibble: 517 x 43
##   system_index area_first_total_ha area_last_total_ha delta_q_ha delta_r_ha
##   <chr>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 0000000000000000~ 77.7          77.7          2.81          5.28
## 2 0000000000000000~ 77.8          77.8         -4.06          5.68
## 3 0000000000000000~ 77.8          77.8        -27.0          29.6
## 4 0000000000000000~ 31.9          31.9          1.31         -0.126
```

```
## 5 0000000000000000~ 32.6 32.6 1.28 -0.0722
## 6 0000000000000000~ 77.9 77.9 6.72 -4.79
## 7 0000000000000000~ 77.9 77.9 5.16 -2.47
## 8 0000000000000000~ 77.8 77.8 -1.96 -2.58
## 9 0000000000000000~ 77.8 77.8 -29.4 26.2
## 10 0000000000000000~ 77.8 77.8 2.37 1.50
## # i 507 more rows
## # i 38 more variables: delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>,
## # lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## # r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## # t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## # last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, geo <chr>,
## # run_id <chr>, years_between.x <dbl>, delta_focal_ha <dbl>, ...
```

```
RF_data_nochange %>% filter(is.na(biogeo))
```

```
## # A tibble: 0 x 43
## # i 43 variables: system_index <chr>, area_first_total_ha <dbl>,
## # area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## # delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## # lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## # r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## # t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year.x <dbl>,
## # n_resurvey <dbl>, plot_id <chr>, geo <chr>, run_id <chr>, ...
```

```
RF_data_nochange %>% filter(first_year.x != first_year.y) # 3 plots
```

```
## # A tibble: 3 x 43
##   system_index area_first_total_ha area_last_total_ha delta_q_ha delta_r_ha
##   <chr>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 0000000000000000~ 77.9 77.9 -2.46 -2.76
## 2 0000000000000000~ 77.9 77.9 -1.58 -4.96
## 3 0000000000000000~ 77.9 77.9 -26.0 8.24
## # i 38 more variables: delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>,
## # lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## # r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## # t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## # last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, geo <chr>,
## # run_id <chr>, years_between.x <dbl>, delta_focal_ha <dbl>,
## # delta_focal_pct <dbl>, time_quartile <chr>, eunis_cambio <lgl>, ...
```

```
RF_data_nochange %>% filter(last_year.x != last_year.y)
```

```
## # A tibble: 0 x 43
## # i 43 variables: system_index <chr>, area_first_total_ha <dbl>,
## # area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## # delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## # lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## # r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## # t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year.x <dbl>,
## # n_resurvey <dbl>, plot_id <chr>, geo <chr>, run_id <chr>, ...
```

```
RF_data_nochange %>% filter(years_between.x != years_between.y) # 3 plots
```

```
## # A tibble: 3 x 43
##   system_index      area_first_total_ha area_last_total_ha delta_q_ha delta_r_ha
##   <chr>                <dbl>                <dbl>        <dbl>        <dbl>
## 1 0000000000000000~      77.9                77.9        -2.46        -2.76
## 2 0000000000000000~      77.9                77.9        -1.58        -4.96
## 3 0000000000000000~      77.9                77.9        -26.0         8.24
## # i 38 more variables: delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>,
## #   lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## #   r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## #   t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## #   last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, geo <chr>,
## #   run_id <chr>, years_between.x <dbl>, delta_focal_ha <dbl>,
## #   delta_focal_pct <dbl>, time_quartile <chr>, eunis_cambio <lgl>, ...
```

Check with my data which first_year is correct!

```
RF_data_change %>% filter(coalesce(biogeo, "NA") != coalesce(BIOGEO, "NA"))
```

```
## # A tibble: 61 x 47
##   system_index      area_first_total_ha area_last_total_ha delta_q_ha delta_r_ha
##   <chr>                <dbl>                <dbl>        <dbl>        <dbl>
## 1 0000000000000000~      77.8                77.8        -5.52         5.06
## 2 0000000000000000~      77.8                77.8       -13.3        -0.708
## 3 0000000000000000~      77.8                77.8        -1.81        -2.00
## 4 0000000000000000~      77.8                77.8        14.4        -20.6
## 5 0000000000000000~      77.8                77.8       -0.103         1.69
## 6 0000000000000000~      77.8                77.8         1.20         0.352
## 7 0000000000000000~      77.8                77.8         1.65        -5.51
## 8 0000000000000000~      77.8                77.8         1.69        -5.56
## 9 0000000000000000~      78.1                78.1        -4.50         3.26
## 10 0000000000000000~      78.2                78.2         1.25         1.06
## # i 51 more rows
## # i 42 more variables: delta_s_ha <dbl>, delta_t_ha <dbl>, eunis_first <chr>,
## #   eunis_last <chr>, lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>,
## #   q_last_ha <dbl>, r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>,
## #   s_last_ha <dbl>, t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>,
## #   first_year.x <dbl>, last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>,
## #   transition <chr>, years_between.x <dbl>, geo <chr>, run_id <chr>, ...
```

```
RF_data_change %>% filter(is.na(biogeo))
```

```
## # A tibble: 0 x 47
## # i 47 variables: system_index <chr>, area_first_total_ha <dbl>,
## #   area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## #   delta_s_ha <dbl>, delta_t_ha <dbl>, eunis_first <chr>, eunis_last <chr>,
## #   lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## #   r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## #   t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## #   last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, transition <chr>, ...
```

```
RF_data_change %>% filter(first_year.x != first_year.y)
```

```
## # A tibble: 0 x 47
## # i 47 variables: system_index <chr>, area_first_total_ha <dbl>,
## #   area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## #   delta_s_ha <dbl>, delta_t_ha <dbl>, eunis_first <chr>, eunis_last <chr>,
## #   lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## #   r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## #   t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## #   last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, transition <chr>, ...
```

```
RF_data_change %>% filter(last_year.x != last_year.y)
```

```
## # A tibble: 0 x 47
## # i 47 variables: system_index <chr>, area_first_total_ha <dbl>,
## #   area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## #   delta_s_ha <dbl>, delta_t_ha <dbl>, eunis_first <chr>, eunis_last <chr>,
## #   lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## #   r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## #   t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## #   last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, transition <chr>, ...
```

```
RF_data_change %>% filter(years_between.x != years_between.y)
```

```
## # A tibble: 0 x 47
## # i 47 variables: system_index <chr>, area_first_total_ha <dbl>,
## #   area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## #   delta_s_ha <dbl>, delta_t_ha <dbl>, eunis_first <chr>, eunis_last <chr>,
## #   lat_updated <dbl>, lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>,
## #   r_first_ha <dbl>, r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>,
## #   t_first_ha <dbl>, t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>,
## #   last_year.x <dbl>, n_resurvey <dbl>, plot_id <chr>, transition <chr>, ...
```

Remove duplicated columns and rename remaining ones

```
RF_data_nochange <- RF_data_nochange %>%
  # TBD for first_year and years_between!
  dplyr::select(-EUNIS1, -BIOGEO, -last_year.y) %>%
  rename(last_year = last_year.x)
```

```
RF_data_change <- RF_data_change %>%
  dplyr::select(-BIOGEO)
```

Join plots with constant habitat and plots with habitat change

```
RF_data_change <- RF_data_change %>%
  dplyr::select(-EUNIS1, -last_year.y) %>%
  # Set eunis1 as "Change" for plots with habitat change
  mutate(eunis1 = "Change") %>%
  rename(last_year = last_year.x)
```

```
RF_data <- bind_rows(RF_data_nochange, RF_data_change)
```

Fix plotIDs in RF_data

Gonzalo used RS_CODE instead of Resurvey project as first part of plot_id.

Retrieve list of correspondences of RS_CODE and Resurvey project from original data:

```
correspondence_list <- initial_dataset %>%
  distinct(RS_CODE, `ReSurvey project`)
```

```
RF_data <- RF_data %>%
  # Add spaces before and after "-"
  mutate(plotID = str_replace_all(plot_id, "-", " - ")) %>%
  # Get RS_CODE: first part of plot_id, before first "-"
  mutate(RS_CODE = sub("-.*", "", plot_id)) %>%
  # Join to get Resurvey project
  left_join(correspondence_list) %>%
  # Construct new plotID
  mutate(plotID = str_replace(plotID, paste0("^", RS_CODE),
    `ReSurvey project`)) %>%
  # Remove unneeded columns
  dplyr::select(-plot_id, -RS_CODE, -`ReSurvey project`) %>%
  # Modify some plotIDs
  mutate(
    plotID = str_replace(
      plotID,
      "^AOPK_HabitatMonitoring - PolickeSteny - Signal -",
      "AOPK_HabitatMonitoring - PolickeSteny-Signal -"
    ),
    plotID = str_replace(
      plotID,
      "^Ecohydrology of Floodplain Meadows - ([0-9]+) - \\1\\s+-\\s+",
      "Ecohydrology of Floodplain Meadows - \\1 - \\1-"
    ),
    plotID = str_replace(
      plotID,
      "^PSET Puy Long - PuyLong - PSET 0 - 1",
      "PSET Puy Long - PuyLong - PSET 0-1"
    ),
    plotID = str_replace(
      plotID,
      "^PSET Puy Long - PuyLong - PSET ([0-9]+) - ([0-9]+)",
      "PSET Puy Long - PuyLong - PSET \\1-\\2"
    ),
    plotID = str_replace(
```

```

    plotID,
    "^Role of AES schemes on the grasslands biodiversity - ([0-9]+) - ([0-9]+) - ([0-9]+) - \\1 - \\2"
    "Role of AES schemes on the grasslands biodiversity - \\1-\\2-\\3 - \\1-\\2-\\3\\4"
  )
)

```

Joining with 'by = join_by(RS_CODE)'

Two types of plot IDs:

```

RF_data <- RF_data %>%
  # Without spaces before and after "-"
  mutate(plot_id = str_replace_all(plotID, " - ", "-")) %>%
  # With spaces before and after "-"
  rename(plot_id_spaced = plotID)

```

Add EUNIS level 2

Useful for some analyses.

```

data_eunis <- initial_dataset %>%
  dplyr::select(PlotObservationID, `ReSurvey project`, RS_CODE, `ReSurvey site`,
    `ReSurvey plot`, year, plotID_AV, EUNIS1, EUNIS2) %>%
  # Remove spaces in plot_id
  mutate(plot_id = str_replace_all(plotID_AV, " - ", "-")) %>%
  # Remove unneeded columns
  dplyr::select(-PlotObservationID, -`ReSurvey project`, -RS_CODE,
    -`ReSurvey site`, -`ReSurvey plot`, -plotID_AV) %>%
  # Keep only plot IDs in RF_data
  filter(plot_id %in% RF_data$plot_id) %>%
  # To check match with EUNIS 1 in RF data
  rename(EUNIS1_check = EUNIS1)

```

```

RF_data <- RF_data %>%
  left_join(data_eunis %>% rename(first_year.x = year,
    EUNIS1_check_first = EUNIS1_check,
    eunis2_first = EUNIS2),
    by = c("plot_id", "first_year.x")) %>%
  left_join(data_eunis %>% rename(last_year = year,
    EUNIS1_check_last = EUNIS1_check,
    eunis2_last = EUNIS2),
    by = c("plot_id", "last_year"))

```

Some checks:

```

RF_data %>% filter(eunis_cambio == F & eunis1 != EUNIS1_check_first)

```

```

## # A tibble: 0 x 50
## # i 50 variables: system_index <chr>, area_first_total_ha <dbl>,
## #   area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,

```

```
## # delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## # lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## # r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## # t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year <dbl>,
## # n_resurvey <dbl>, geo <chr>, run_id <chr>, years_between.x <dbl>, ...
```

```
RF_data %>% filter(eunis_cambio == F & eunis1 != EUNIS1_check_last)
```

```
## # A tibble: 0 x 50
## # i 50 variables: system_index <chr>, area_first_total_ha <dbl>,
## # area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## # delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## # lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## # r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## # t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year <dbl>,
## # n_resurvey <dbl>, geo <chr>, run_id <chr>, years_between.x <dbl>, ...
```

```
RF_data %>% filter(eunis_cambio == T & eunis_first != EUNIS1_check_first)
```

```
## # A tibble: 0 x 50
## # i 50 variables: system_index <chr>, area_first_total_ha <dbl>,
## # area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## # delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## # lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## # r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## # t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year <dbl>,
## # n_resurvey <dbl>, geo <chr>, run_id <chr>, years_between.x <dbl>, ...
```

```
RF_data %>% filter(eunis_cambio == T & eunis_last != EUNIS1_check_last)
```

```
## # A tibble: 0 x 50
## # i 50 variables: system_index <chr>, area_first_total_ha <dbl>,
## # area_last_total_ha <dbl>, delta_q_ha <dbl>, delta_r_ha <dbl>,
## # delta_s_ha <dbl>, delta_t_ha <dbl>, eunis1 <chr>, lat_updated <dbl>,
## # lon_updated <dbl>, q_first_ha <dbl>, q_last_ha <dbl>, r_first_ha <dbl>,
## # r_last_ha <dbl>, s_first_ha <dbl>, s_last_ha <dbl>, t_first_ha <dbl>,
## # t_last_ha <dbl>, biogeo <chr>, first_year.x <dbl>, last_year <dbl>,
## # n_resurvey <dbl>, geo <chr>, run_id <chr>, years_between.x <dbl>, ...
```

All OK.

```
RF_data <- RF_data %>%
  # Remove unneeded columns
  dplyr::select(-EUNIS1_check_first, -EUNIS1_check_last) %>%
  # create eunis2 column
  mutate(eunis2 = case_when(
    eunis1 == "Change" ~ "Change",
    eunis2_first == eunis2_last ~ eunis2_first,
    !is.na(eunis2_first) & is.na(eunis2_last) ~ eunis2_first,
    is.na(eunis2_first) & !is.na(eunis2_last) ~ eunis2_last,
    eunis2_first != eunis2_last ~ "Change"
  ))
```

Save as .rds

```
saveRDS(RF_data, here("data", "clean", "RF_data.rds"))
```

Change in extent

Plots with constant habitat

Change in extent by habitat type

```
mod_extent_eunis1_noch <- lm(delta_focal_pct ~ eunis1,  
                             data = RF_data %>% filter(eunis_cambio == F))  
Anova(mod_extent_eunis1_noch)
```

```
## Anova Table (Type II tests)  
##  
## Response: delta_focal_pct  
##           Sum Sq   Df F value    Pr(>F)  
## eunis1      10463    3  28.206 < 2.2e-16 ***  
## Residuals 1391013 11250  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_extent_eunis1_noch, pairwise ~ eunis1)
```

```
## $emmeans  
## eunis1 emmean    SE    df lower.CL upper.CL  
## Q      -2.493 0.427 11250 -3.3302  -1.656  
## R       1.255 0.137 11250  0.9868   1.522  
## S       2.201 0.405 11250  1.4057   2.995  
## T       0.482 0.197 11250  0.0964   0.867  
##  
## Confidence level used: 0.95  
##  
## $contrasts  
## contrast estimate    SE    df t.ratio p.value  
## Q - R      -3.748 0.448 11250 -8.359 <0.0001  
## Q - S      -4.694 0.589 11250 -7.970 <0.0001  
## Q - T      -2.975 0.470 11250 -6.328 <0.0001  
## R - S      -0.946 0.428 11250 -2.211  0.1204  
## R - T       0.773 0.239 11250  3.229  0.0068  
## S - T       1.719 0.451 11250  3.815  0.0008  
##  
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_extent_eunis1_noch)$r.squared
```

```
## [1] 0.007465366
```

Change in extent by number of years between first and last survey

```
mod_extent_yrs_eunis1_noch <- lm(abs(delta_focal_pct) ~ years_between.x * eunis1,  
  data = RF_data %>% filter(eunis_cambio == F))  
Anova(mod_extent_yrs_eunis1_noch)
```

```
## Anova Table (Type II tests)  
##  
## Response: abs(delta_focal_pct)  
##  
##           Sum Sq   Df F value    Pr(>F)  
## years_between.x    4368     1  63.8104 1.503e-15 ***  
## eunis1             61214     3 298.0648 < 2.2e-16 ***  
## years_between.x:eunis1  1516     3   7.3823 6.141e-05 ***  
## Residuals          769870 11246  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(mod_extent_yrs_eunis1_noch)
```

```
##  
## Call:  
## lm(formula = abs(delta_focal_pct) ~ years_between.x * eunis1,  
##     data = RF_data %>% filter(eunis_cambio == F))  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -11.666  -5.150  -2.071    2.135   69.436   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    6.82332    0.75614   9.024 < 2e-16 ***  
## years_between.x  0.41130    0.11212   3.668 0.000245 ***  
## eunis1R         0.98936    0.77438   1.278 0.201413      
## eunis1S        -3.49695    1.06192  -3.293 0.000994 ***  
## eunis1T        -3.89323    0.82305  -4.730 2.27e-06 ***  
## years_between.x:eunis1R -0.30107    0.11337  -2.656 0.007926 **  
## years_between.x:eunis1S  0.09868    0.15485   0.637 0.523943      
## years_between.x:eunis1T -0.33294    0.11630  -2.863 0.004209 **  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 8.274 on 11246 degrees of freedom  
## Multiple R-squared:  0.07656,    Adjusted R-squared:  0.07598   
## F-statistic: 133.2 on 7 and 11246 DF,  p-value: < 2.2e-16
```

```
emtrends(mod_extent_yrs_eunis1_noch, pairwise ~ eunis1, var = "years_between.x")
```

```
## $emtrends
## eunis1 years_between.x.trend SE df lower.CL upper.CL
## Q 0.4113 0.1120 11246 0.1915 0.631
## R 0.1102 0.0168 11246 0.0774 0.143
## S 0.5100 0.1070 11246 0.3006 0.719
## T 0.0784 0.0309 11246 0.0178 0.139
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## Q - R 0.3011 0.1130 11246 2.656 0.0396
## Q - S -0.0987 0.1550 11246 -0.637 0.9200
## Q - T 0.3329 0.1160 11246 2.863 0.0219
## R - S -0.3998 0.1080 11246 -3.698 0.0012
## R - T 0.0319 0.0352 11246 0.907 0.8014
## S - T 0.4316 0.1110 11246 3.882 0.0006
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

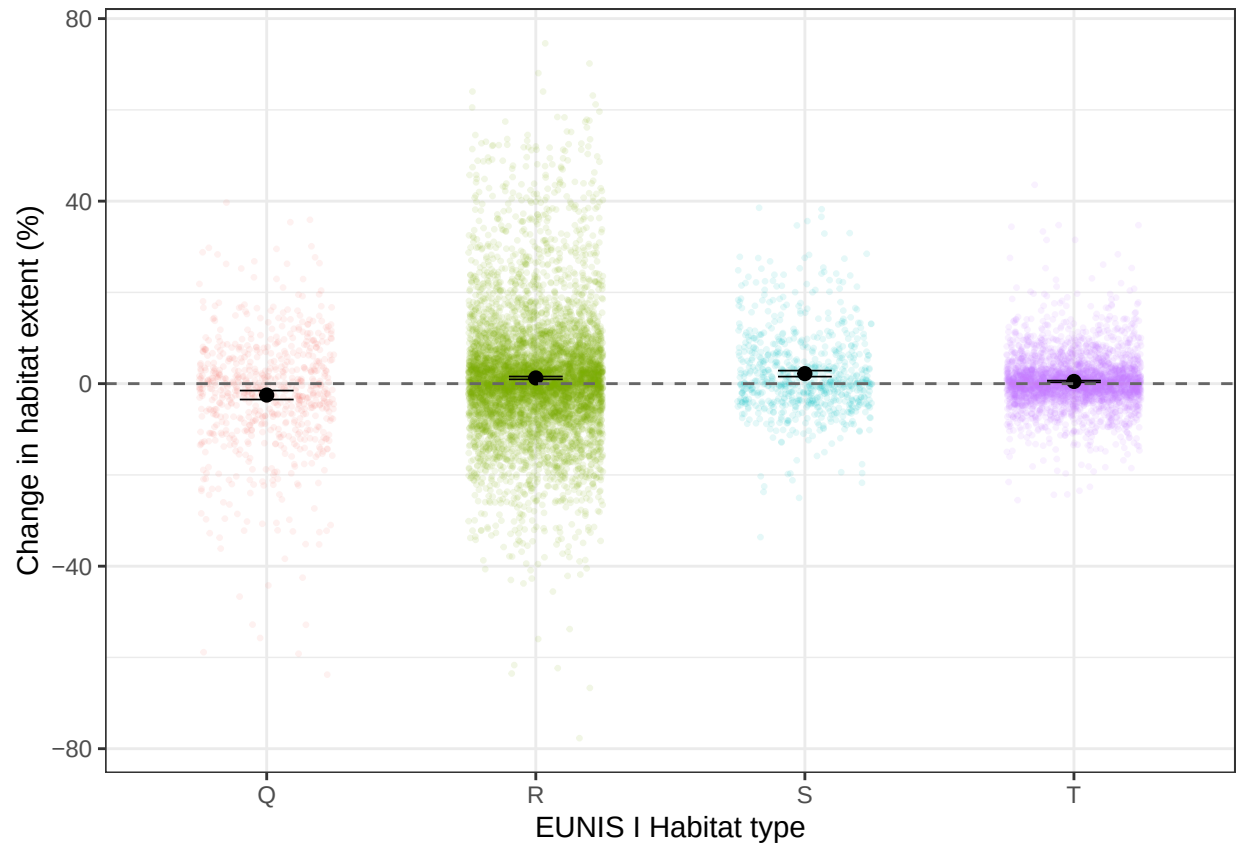
Proportion of variance explained:

```
summary(mod_extent_yrs_eunis1_noch)$r.squared
```

```
## [1] 0.07655564
```

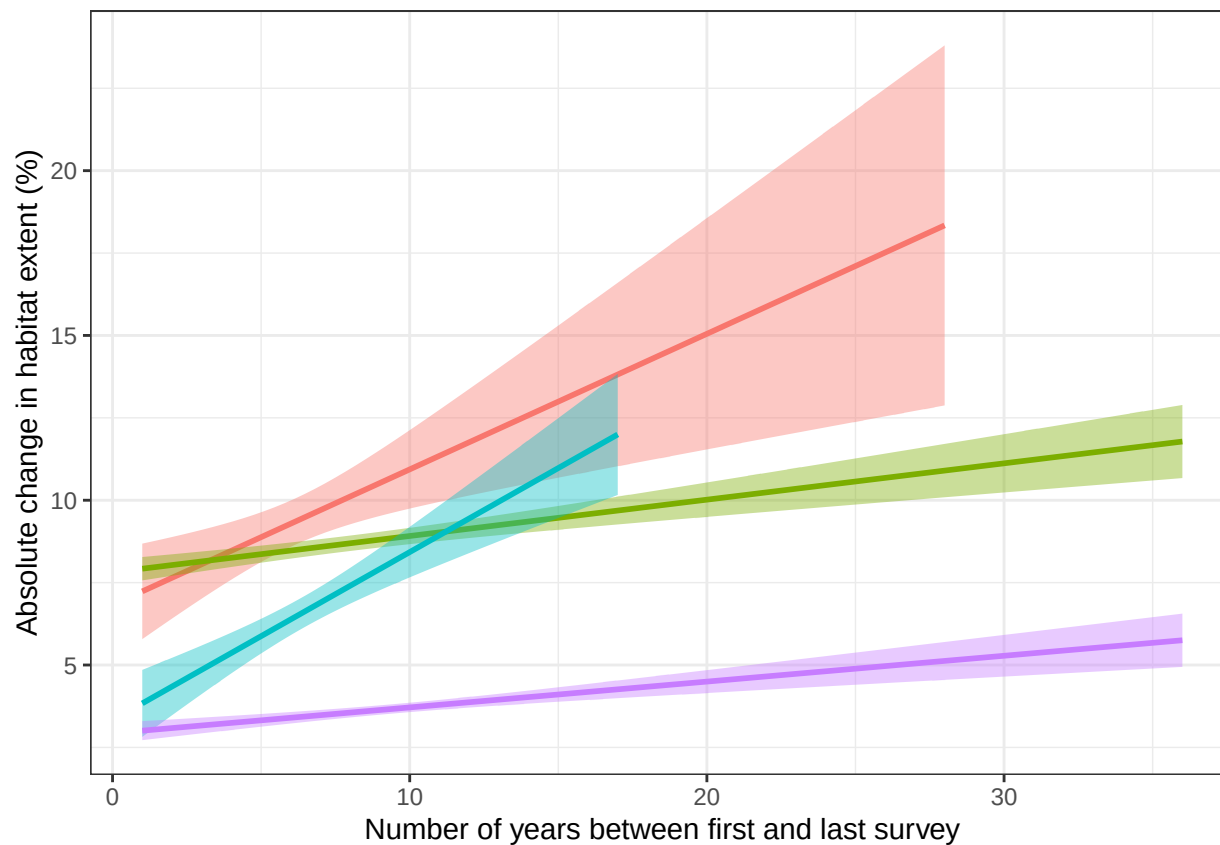
Figure 1

```
figla <- ggplot(RF_data %>% filter(eunis_cambio == F),
  aes(x = eunis1, y = delta_focal_pct, fill = eunis1)) +
  #geom_half_violin(side = "r", alpha = 0.8, position = position_nudge(x = 0.2)) +
  geom_point(aes(color = eunis1),
    position = position_jitter(width = 0.25), size = 0.5, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  # geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  # geom_text(data = cld(emmeans(mod_extent_eunis1_noch, ~ eunis1),
  #   Letters = letters) %>%
  #   as.data.frame(),
  #   aes(x = eunis1, y = 75, label = .group),
  #   inherit.aes = FALSE, size = 5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  labs(x = "EUNIS I Habitat type", y = "Change in habitat extent (%)") +
  theme_bw() +
  theme(legend.position = "none")
figla
```



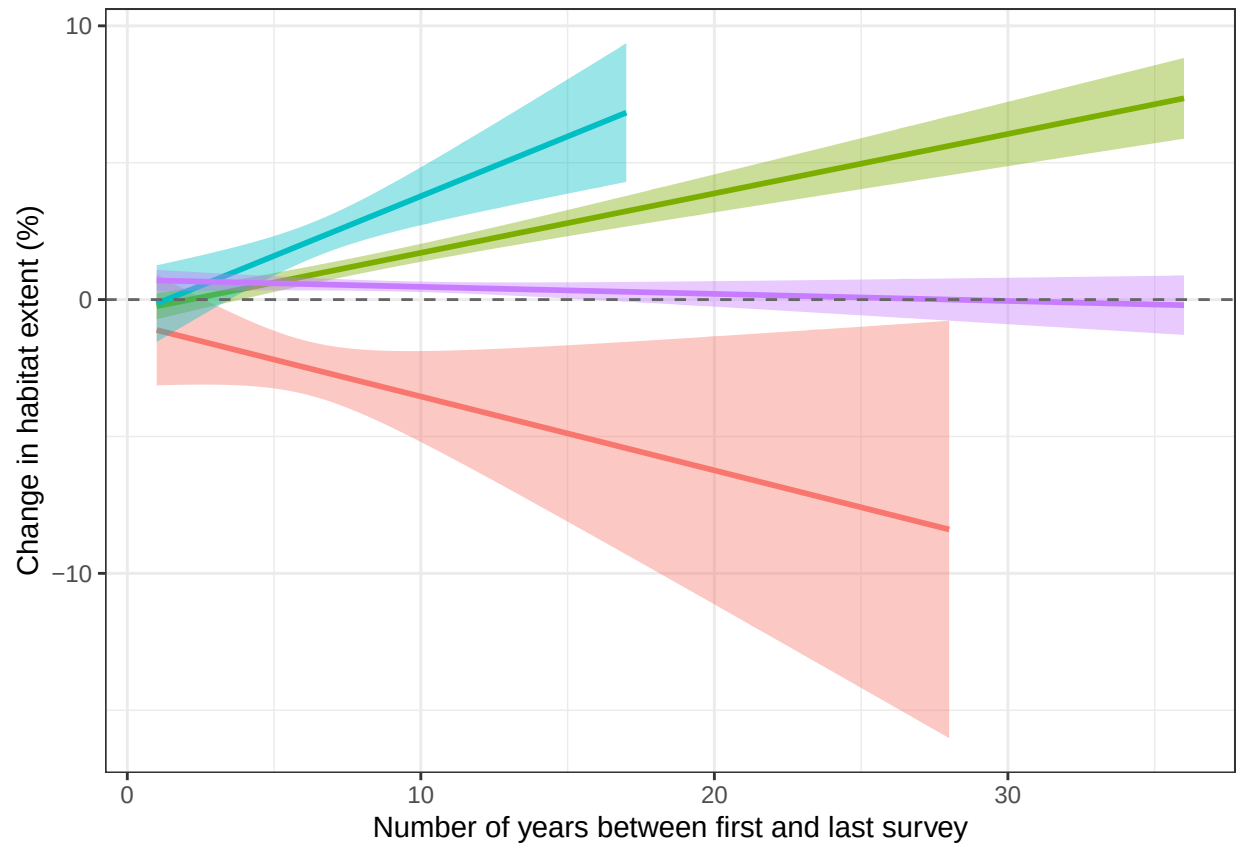
```
fig1b <- ggplot(RF_data %>% filter(eunis_cambio == F),
               aes(x = years_between.x, y = abs(delta_focal_pct),
                   color = eunis1, fill = eunis1, group = eunis1)) +
  # geom_point(size = 1, alpha = 0.1) +
  geom_smooth(method = "lm") +
  labs(x = "Number of years between first and last survey",
       y = "Absolute change in habitat extent (%)") +
  theme_bw() +
  theme(legend.position = "none")
fig1b
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



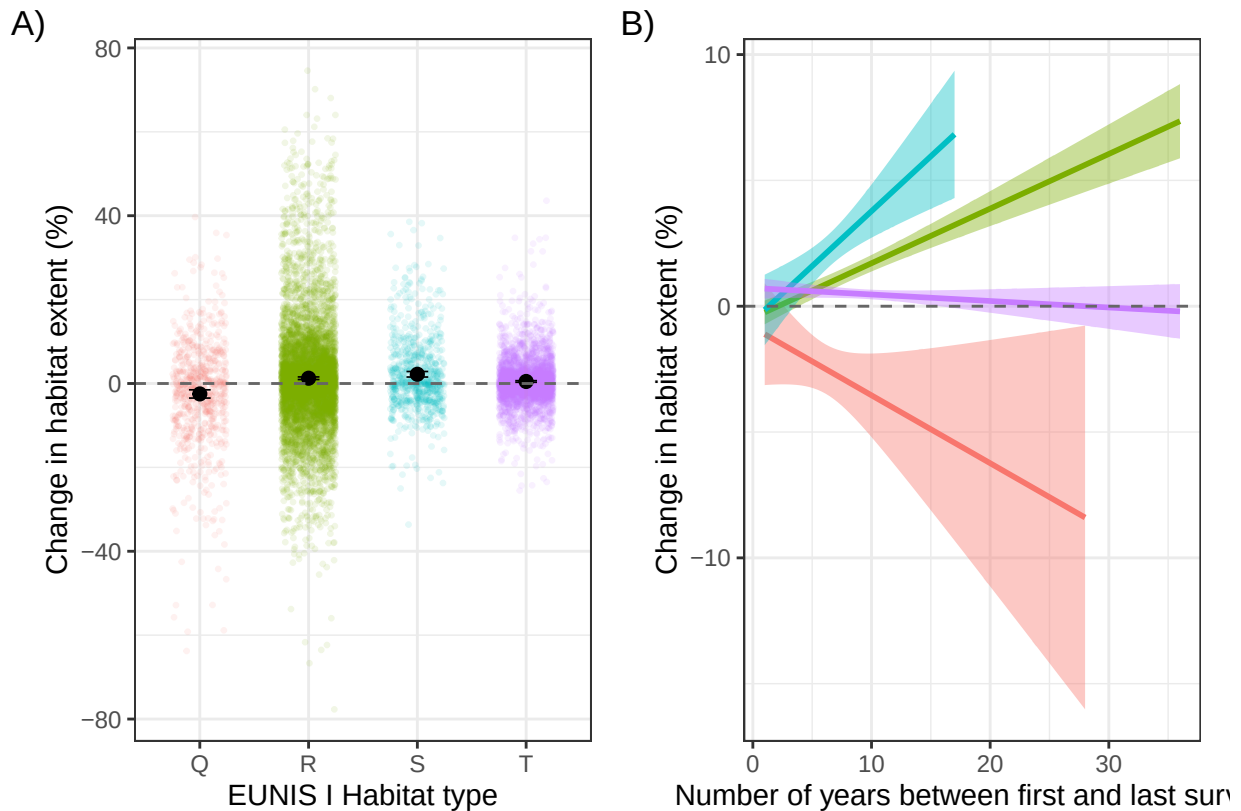
```
fig1b_alt <- ggplot(RF_data %>% filter(eunis_cambio == F),
  aes(x = years_between.x, y = delta_focal_pct,
    color = eunis1, fill = eunis1, group = eunis1)) +
  # geom_point(size = 1, alpha = 0.1) +
  geom_smooth(method = "lm") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  labs(x = "Number of years between first and last survey",
    y = "Change in habitat extent (%)") +
  theme_bw() +
  theme(legend.position = "none")
fig1b_alt
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
fig1 <- fig1a + fig1b_alt +
  plot_layout(ncol = 2) + plot_annotation(tag_levels = list(c("A"), "B"))
fig1
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggsave(fig1, filename = "output/figures/fig1.jpeg",
       width = 24, height = 12, units = "cm", dpi = 300)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Plots with habitat change

Change in extent by habitat type

```
mod_extent_eunis_first_ch <- lm(delta_focal_pct ~ eunis_first,
                                data = RF_data %>% filter(eunis_cambio == T))
Anova(mod_extent_eunis_first_ch)
```

```
## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value Pr(>F)
## eunis_first  2005  3  3.4789 0.01653 *
## Residuals   50148 261
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_extent_eunis_first_ch, pairwise ~ eunis_first)
```

```
## $emmeans
##   eunis_first emmean   SE  df lower.CL upper.CL
##   Q           -6.674 1.41 261   -9.46   -3.8882
##   R           -2.606 1.37 261   -5.30    0.0833
##   S            0.158 1.90 261   -3.59    3.9069
##   T            0.906 3.84 261   -6.66    8.4766
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast estimate   SE  df t.ratio p.value
##   Q - R      -4.068 1.97 261   -2.069  0.1662
##   Q - S      -6.832 2.37 261   -2.880  0.0222
##   Q - T      -7.580 4.10 261   -1.850  0.2523
##   R - S      -2.764 2.34 261   -1.180  0.6403
##   R - T      -3.513 4.08 261   -0.861  0.8249
##   S - T      -0.749 4.29 261   -0.175  0.9981
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

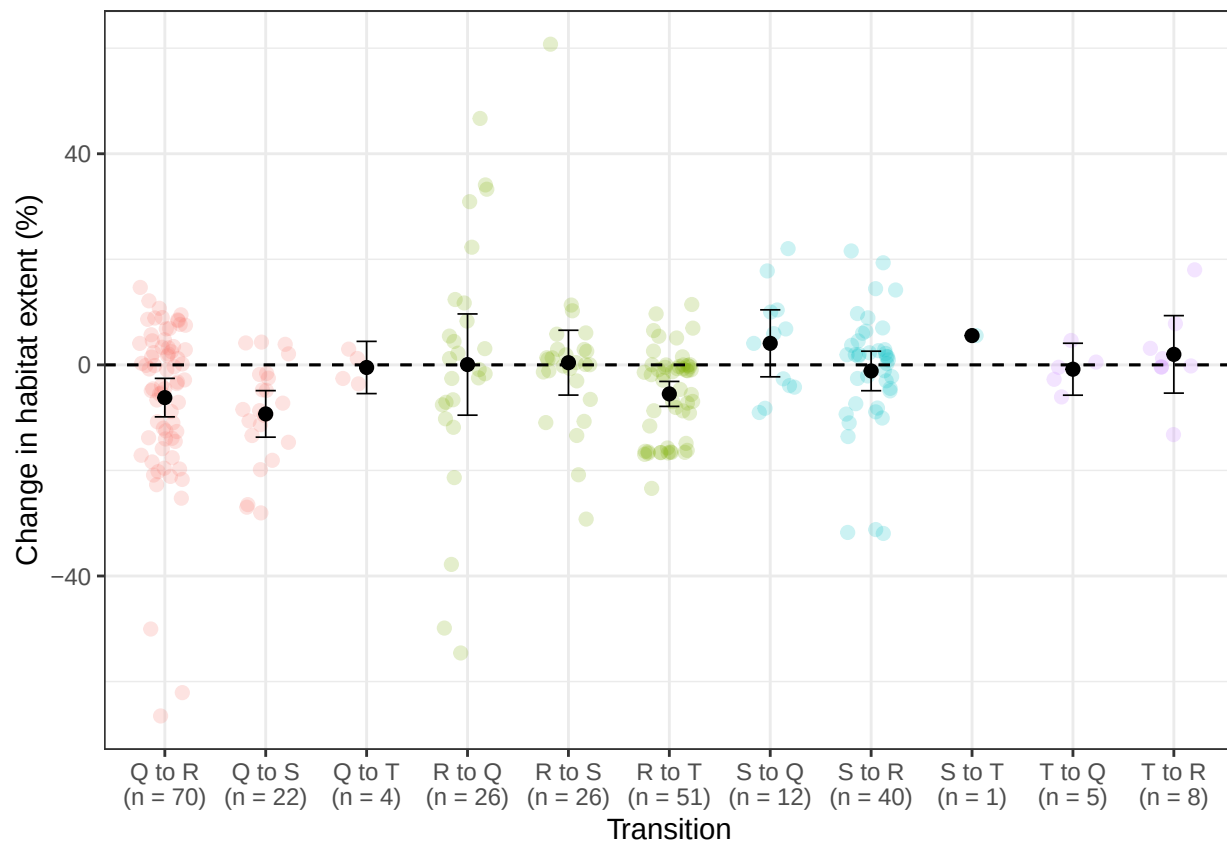
Proportion of variance explained:

```
summary(mod_extent_eunis_first_ch)$r.squared
```

```
## [1] 0.03844985
```

Change in extent by transition type

```
ggplot(RF_data %>% filter(eunis_cambio == T) %>%
  left_join(RF_data %>% filter(eunis_cambio == TRUE) %>%
    count(transition),
    by = "transition") %>%
  mutate(transition_label = paste0(transition, "\n(n = ", n, ")")),
  aes(x = transition_label, y = delta_focal_pct, fill = eunis_first)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis_first),
    position = position_jitter(width = 0.25), size = 2, alpha = 0.2) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  labs(x = "Transition", y = "Change in habitat extent (%)") +
  theme_bw() +
  theme(legend.position = "none")
```



Change in extent by number of years between first and last survey

```
mod_extent_yrs_eunis_first_ch <- lm(abs(delta_focal_pct) ~
                                     years_between.x * eunis_first,
                                     data = RF_data %>% filter(eunis_cambio == T))
Anova(mod_extent_yrs_eunis_first_ch)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##
##               Sum Sq Df F value Pr(>F)
## years_between.x      78.5  1  0.6663 0.4151
## eunis_first         564.8  3  1.5985 0.1902
## years_between.x:eunis_first 467.4  3  1.3229 0.2674
## Residuals          30269.1 257
```

```
summary(mod_extent_yrs_eunis_first_ch)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x * eunis_first,
##     data = RF_data %>% filter(eunis_cambio == T))
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.707  -7.231  -2.509   2.699  56.113
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.8645     2.8478   2.762  0.00617 **
## years_between.x    0.4242     0.4291   0.989  0.32378
## eunis_firstR      2.8984     3.4450   0.841  0.40095
## eunis_firstS      4.6491     4.0822   1.139  0.25581
## eunis_firstT     -7.2564     7.8480  -0.925  0.35604
## years_between.x:eunis_firstR -0.4943     0.4475  -1.105  0.27035
## years_between.x:eunis_firstS -0.9609     0.5290  -1.816  0.07048 .
## years_between.x:eunis_firstT  0.3964     1.4619   0.271  0.78649
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.85 on 257 degrees of freedom
## Multiple R-squared:  0.03405,    Adjusted R-squared:  0.007739
## F-statistic: 1.294 on 7 and 257 DF,  p-value: 0.2534
```

```
mod_extent_yrs_ch <- lm(abs(delta_focal_pct) ~ years_between.x ,
                        data = RF_data %>% filter(eunis_cambio == T))
Anova(mod_extent_yrs_ch)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##              Sum Sq Df F value Pr(>F)
## years_between.x    34.7  1  0.2919 0.5895
## Residuals        31301.3 263
```

```
summary(mod_extent_yrs_ch)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == T))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.684  -7.185  -3.132   2.793  56.881
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.96802     1.12944   8.826  <2e-16 ***
## years_between.x -0.05446     0.10080  -0.540   0.589
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.91 on 263 degrees of freedom
## Multiple R-squared:  0.001109,    Adjusted R-squared:  -0.00269
## F-statistic: 0.2919 on 1 and 263 DF,  p-value: 0.5895
```

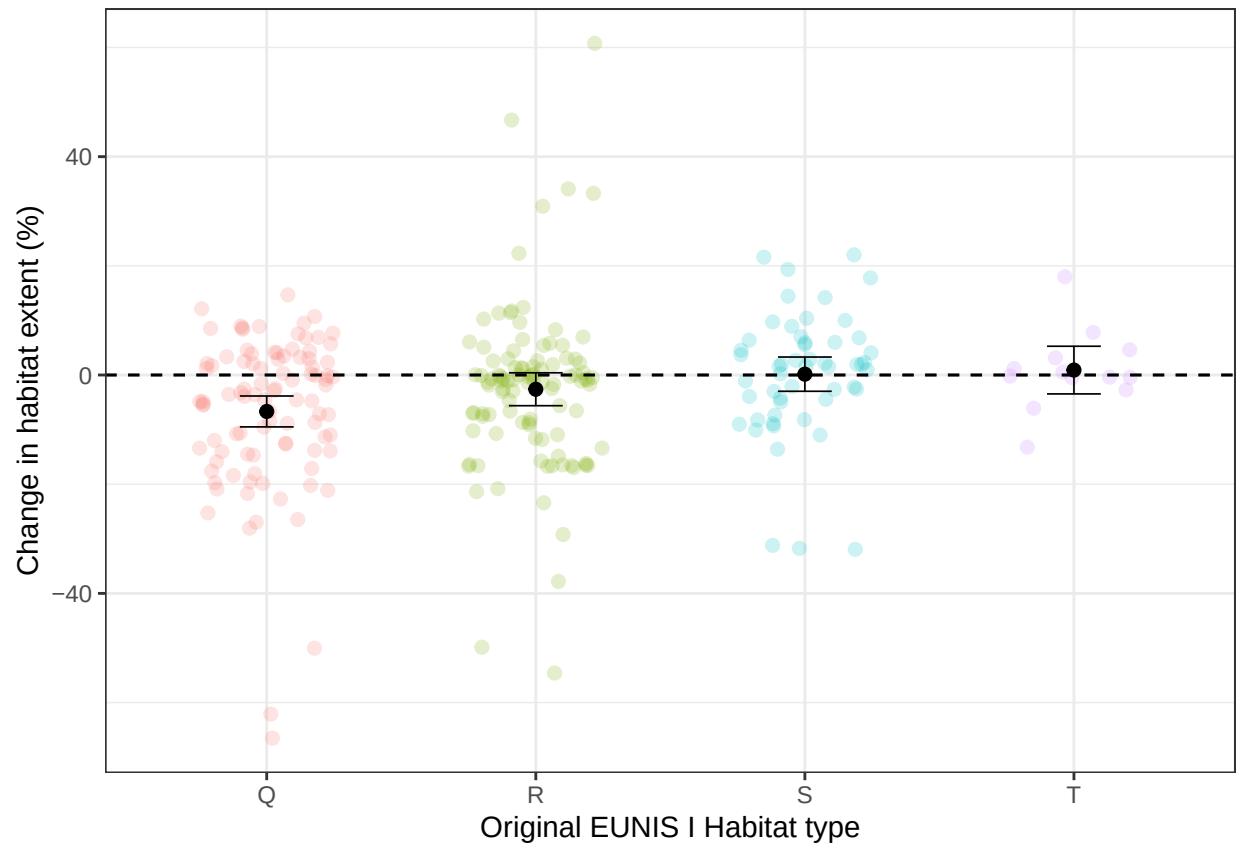
Proportion of variance explained:

```
summary(mod_extent_yrs_eunis_first_ch)$r.squared
```

```
## [1] 0.03404847
```

Appendix 1

```
app1 <- ggplot(RF_data %>% filter(eunis_cambio == T),
               aes(x = eunis_first, y = delta_focal_pct, fill = eunis_first)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  #geom_half_violin(side = "r", alpha = 0.8, position = position_nudge(x = 0.2)) +
  geom_point(aes(color = eunis_first),
             position = position_jitter(width = 0.25), size = 2, alpha = 0.2) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
              linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
              color = "black", fill = "black") +
  # geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  # geom_text(data = cld(emmeans(mod_extent_eunis1_ch, ~ eunis_first),
  #                           Letters = letters) %>%
  #           as.data.frame(),
  #           aes(x = eunis_first, y = 75, label = .group),
  #           inherit.aes = FALSE, size = 5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  labs(x = "Original EUNIS I Habitat type", y = "Change in habitat extent (%)") +
  theme_bw() +
  theme(legend.position = "none")
app1
```



```
ggsave(app1, filename = "output/figures/app1.jpeg",
        width = 12, height = 12, units = "cm", dpi = 300)
```

Change in extent by bioregion

Plots with constant habitat

```
table((RF_data %>% filter(eunis_cambio == F))$eunis1,
      (RF_data %>% filter(eunis_cambio == F))$biogeo)
```

```
##
##      ALP  ATL  CON  MED  PAN
##  Q      9  172  497    0    0
##  R    412 1013 4834   122  242
##  S      2  505  245    0    0
##  T     18  233 2856    0   94
```

Change in extent by habitat type

```

mod_extent_eunis1_noch_ALP <- lm(delta_focal_pct ~ eunis1,
                                data = RF_data %>% filter(eunis_cambio == F) %>%
                                filter(biogeo == "ALP"))
mod_extent_eunis1_noch_ATL <- lm(delta_focal_pct ~ eunis1,
                                data = RF_data %>% filter(eunis_cambio == F) %>%
                                filter(biogeo == "ATL"))
mod_extent_eunis1_noch_CON <- lm(delta_focal_pct ~ eunis1,
                                data = RF_data %>% filter(eunis_cambio == F) %>%
                                filter(biogeo == "CON"))
# Not possible in MED because all are R
mod_extent_eunis1_noch_PAN <- lm(delta_focal_pct ~ eunis1,
                                data = RF_data %>% filter(eunis_cambio == F) %>%
                                filter(biogeo == "PAN"))

Anova(mod_extent_eunis1_noch_ALP)

```

```

## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value Pr(>F)
## eunis1      592   3  1.3359 0.2621
## Residuals 64520 437

```

```
Anova(mod_extent_eunis1_noch_ATL)
```

```

## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value    Pr(>F)
## eunis1      7817   3  26.067 < 2.2e-16 ***
## Residuals 191830 1919
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_extent_eunis1_noch_CON)
```

```

## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value    Pr(>F)
## eunis1      5106   3  12.997 1.819e-08 ***
## Residuals 1103771 8428
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_extent_eunis1_noch_PAN)
```

```

## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value Pr(>F)
## eunis1       10.4   1  0.1979 0.6567
## Residuals 17555.3 334

```

```
emmeans(mod_extent_eunis1_noch_ATL, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1 emmean SE df lower.CL upper.CL
## Q -5.03 0.762 1919 -6.520 -3.53
## R 1.47 0.314 1919 0.855 2.09
## S 2.62 0.445 1919 1.743 3.49
## T 1.73 0.655 1919 0.449 3.02
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## Q - R -6.496 0.825 1919 -7.878 <0.0001
## Q - S -7.641 0.883 1919 -8.656 <0.0001
## Q - T -6.759 1.010 1919 -6.725 <0.0001
## R - S -1.145 0.545 1919 -2.102 0.1527
## R - T -0.263 0.726 1919 -0.362 0.9838
## S - T 0.882 0.792 1919 1.114 0.6809
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_extent_eunis1_noch_CON, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1 emmean SE df lower.CL upper.CL
## Q -1.660 0.513 8428 -2.6665 -0.654
## R 1.297 0.165 8428 0.9740 1.619
## S 1.393 0.731 8428 -0.0403 2.826
## T 0.262 0.214 8428 -0.1573 0.682
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## Q - R -2.9568 0.539 8428 -5.485 <0.0001
## Q - S -3.0531 0.893 8428 -3.418 0.0035
## Q - T -1.9226 0.556 8428 -3.457 0.0031
## R - S -0.0963 0.749 8428 -0.128 0.9992
## R - T 1.0342 0.270 8428 3.829 0.0007
## S - T 1.1305 0.762 8428 1.484 0.4473
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_extent_eunis1_noch_ALP)$r.squared
```

```
## [1] 0.009087771
```

```
summary(mod_extent_eunis1_noch_ATL)$r.squared
```

```
## [1] 0.03915549
```

```
summary(mod_extent_eunis1_noch_CON)$r.squared
```

```
## [1] 0.004605064
```

```
summary(mod_extent_eunis1_noch_PAN)$r.squared
```

```
## [1] 0.0005922194
```

Change in extent by number of years between first and last survey

```
mod_extent_yrs_noch_ALP <- lm(abs(delta_focal_pct) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeo == "ALP"))  
mod_extent_yrs_noch_ATL <- lm(abs(delta_focal_pct) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeo == "ATL"))  
mod_extent_yrs_noch_CON <- lm(abs(delta_focal_pct) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeo == "CON"))  
mod_extent_yrs_noch_MED <- lm(abs(delta_focal_pct) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeo == "MED"))  
mod_extent_yrs_noch_PAN <- lm(abs(delta_focal_pct) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeo == "PAN"))  
Anova(mod_extent_yrs_noch_ALP)
```

```
## Anova Table (Type II tests)  
##  
## Response: abs(delta_focal_pct)  
##           Sum Sq Df F value Pr(>F)  
## years_between.x    101  1  1.3161 0.2519  
## Residuals       33814 439
```

```
Anova(mod_extent_yrs_noch_ATL)
```

```
## Anova Table (Type II tests)  
##  
## Response: abs(delta_focal_pct)  
##           Sum Sq Df F value Pr(>F)  
## years_between.x     80  1  1.3287 0.2492  
## Residuals       115398 1921
```

```
Anova(mod_extent_yrs_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x    737     1  9.3944 0.002183 **
## Residuals       660943 8430
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_extent_yrs_noch_MED)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 4483.7     1 429.08 < 2.2e-16 ***
## Residuals       1253.9 120
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_extent_yrs_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x   48.6     1  1.3179 0.2518
## Residuals       12322.0 334
```

```
summary(mod_extent_yrs_noch_ALP)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ALP"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.518 -6.130 -2.413  2.739 45.633
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.85839    0.64047  12.270  <2e-16 ***
## years_between.x 0.04929    0.04296   1.147    0.252
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.776 on 439 degrees of freedom
## Multiple R-squared:  0.002989, Adjusted R-squared:  0.0007179
## F-statistic: 1.316 on 1 and 439 DF, p-value: 0.2519
```

```
summary(mod_extent_yrs_noch_ATL)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ATL"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.261 -5.355 -2.426  2.004 49.057
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.40469    0.33142  19.325  <2e-16 ***
## years_between.x  0.03967    0.03441   1.153   0.249
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.751 on 1921 degrees of freedom
## Multiple R-squared:  0.0006912, Adjusted R-squared:  0.000171
## F-statistic: 1.329 on 1 and 1921 DF, p-value: 0.2492
```

```
summary(mod_extent_yrs_noch_CON)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "CON"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.648 -5.576 -3.080  2.207 70.589
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.87044    0.17654  38.918 < 2e-16 ***
## years_between.x  0.05765    0.01881   3.065  0.00218 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.855 on 8430 degrees of freedom
## Multiple R-squared:  0.001113, Adjusted R-squared:  0.0009947
## F-statistic: 9.394 on 1 and 8430 DF, p-value: 0.002183
```

```
summary(mod_extent_yrs_noch_MED)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "MED"))
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.7014  -0.7063  -0.1792   0.5054  17.0908
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.4467     0.3987  -6.137 1.12e-08 ***
## years_between.x    3.1529     0.1522  20.714 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.233 on 120 degrees of freedom
## Multiple R-squared:  0.7815, Adjusted R-squared:  0.7796
## F-statistic: 429.1 on 1 and 120 DF,  p-value: < 2.2e-16
```

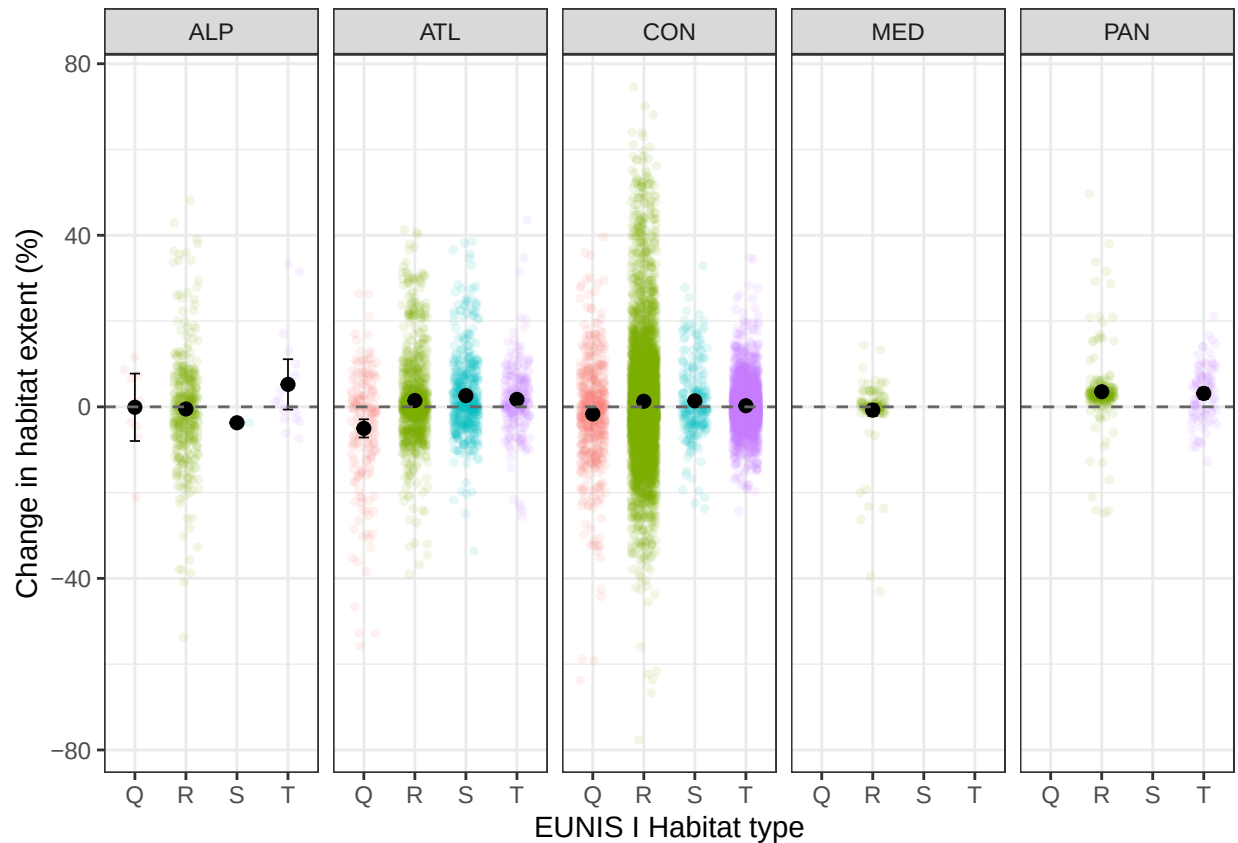
```
summary(mod_extent_yrs_noch_PAN)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeom == "PAN"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.486  -2.809  -1.923   0.062  43.829
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     6.34942     1.04996   6.047 3.94e-09 ***
## years_between.x -0.09023     0.07860  -1.148   0.252
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.074 on 334 degrees of freedom
## Multiple R-squared:  0.00393, Adjusted R-squared:  0.0009482
## F-statistic: 1.318 on 1 and 334 DF,  p-value: 0.2518
```

Appendix 2

```
app2 <- ggplot(RF_data %>% filter(eunis_cambio == F),
  aes(x = eunis1, y = delta_focal_pct, fill = eunis1)) +
  #geom_half_violin(side = "r", alpha = 0.8, position = position_nudge(x = 0.2)) +
  geom_point(aes(color = eunis1),
    position = position_jitter(width = 0.25), size = 1, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  # geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  # geom_text(data = cld(emmeans(mod_extent_eunis1_noch, ~ eunis1),
  #   Letters = letters) %>%
```

```
#           as.data.frame(),
#           aes(x = eunis1, y = 75, label = .group),
#           inherit.aes = FALSE, size = 5) +
geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
facet_wrap(~ biogeo, ncol = 5) +
labs(x = "EUNIS I Habitat type", y = "Change in habitat extent (%)") +
theme_bw() +
theme(legend.position = "none")
app2
```



```
ggsave(app2, filename = "output/figures/app2.jpeg",
        width = 30, height = 12, units = "cm", dpi = 300)
```

Plots with habitat change

```
table((RF_data %>% filter(eunis_cambio == T))$eunis_first,
      (RF_data %>% filter(eunis_cambio == T))$biogeo)
```

```
##
##      ALP ATL CON PAN
##  Q    7  28  61   0
##  R    9  20  73   1
```

```
##   S   0  24  29   0
##   T   4   0   9   0
```

Change in extent by habitat type

```
mod_extent_eunis_first_ch_ALP <- lm(delta_focal_pct ~ eunis_first,
                                     data = RF_data %>%
                                     filter(eunis_cambio == T) %>%
                                     filter(biogeo == "ALP"))
mod_extent_eunis_first_ch_ATL <- lm(delta_focal_pct ~ eunis_first,
                                     data = RF_data %>%
                                     filter(eunis_cambio == T) %>%
                                     filter(biogeo == "ATL"))
mod_extent_eunis_first_ch_CON <- lm(delta_focal_pct ~ eunis_first,
                                     data = RF_data %>%
                                     filter(eunis_cambio == T) %>%
                                     filter(biogeo == "CON"))
Anova(mod_extent_eunis_first_ch_ALP)
```

```
## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value Pr(>F)
## eunis_first  167.9  2  0.4708 0.6324
## Residuals    3031.1 17
```

```
Anova(mod_extent_eunis_first_ch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value    Pr(>F)
## eunis_first 1659.9  2  8.4151 0.0005367 ***
## Residuals    6805.4 69
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_extent_eunis_first_ch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: delta_focal_pct
##           Sum Sq Df F value Pr(>F)
## eunis_first    643  3  0.9054 0.4398
## Residuals   39760 168
```

```
emmeans(mod_extent_eunis_first_ch_ATL, pairwise ~ eunis_first)
```

```
## $emmeans
```

```
## eunis_first emmean SE df lower.CL upper.CL
## Q -8.506 1.88 69 -12.25 -4.76
## R -0.109 2.22 69 -4.54 4.32
## S 2.227 2.03 69 -1.82 6.27
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## Q - R -8.40 2.91 69 -2.888 0.0141
## Q - S -10.73 2.76 69 -3.885 0.0007
## R - S -2.34 3.01 69 -0.777 0.7185
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Proportion of variance explained:

```
summary(mod_extent_eunis_first_ch_ALP)$r.squared
```

```
## [1] 0.05248643
```

```
summary(mod_extent_eunis_first_ch_ATL)$r.squared
```

```
## [1] 0.1960864
```

```
summary(mod_extent_eunis_first_ch_CON)$r.squared
```

```
## [1] 0.01591136
```

Change in extent by number of years between first and last survey

```
mod_extent_yrs_ch_ALP <- lm(abs(delta_focal_pct) ~ years_between.x,
  data = RF_data %>% filter(eunis_cambio == T) %>%
  filter(biogeo == "ALP"))
mod_extent_yrs_ch_ATL <- lm(abs(delta_focal_pct) ~ years_between.x,
  data = RF_data %>% filter(eunis_cambio == T) %>%
  filter(biogeo == "ATL"))
mod_extent_yrs_ch_CON <- lm(abs(delta_focal_pct) ~ years_between.x,
  data = RF_data %>% filter(eunis_cambio == T) %>%
  filter(biogeo == "CON"))
Anova(mod_extent_yrs_ch_ALP)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
## Sum Sq Df F value Pr(>F)
## years_between.x 15.45 1 0.1552 0.6982
## Residuals 1791.25 18
```

```
Anova(mod_extent_yrs_ch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##           Sum Sq Df F value  Pr(>F)
## years_between.x 171.8  1  3.0074 0.08729 .
## Residuals      3999.0 70
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_extent_yrs_ch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: abs(delta_focal_pct)
##           Sum Sq Df F value Pr(>F)
## years_between.x   19.9  1  0.1353 0.7134
## Residuals      25036.5 170
```

```
summary(mod_extent_yrs_ch_ALP)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == T) %>% filter(biogeo == "ALP"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.734 -6.269 -4.273  5.645 29.576
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.6288     3.6730   2.077   0.0524 .
## years_between.x  0.1009     0.2560   0.394   0.6982
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.976 on 18 degrees of freedom
## Multiple R-squared:  0.008551, Adjusted R-squared: -0.04653
## F-statistic: 0.1552 on 1 and 18 DF, p-value: 0.6982
```

```
summary(mod_extent_yrs_ch_ATL)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == T) %>% filter(biogeo == "ATL"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -9.343 -5.353 -2.000 3.195 24.209
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.7697     1.7548   6.137 4.48e-08 ***
## years_between.x -0.3402     0.1961  -1.734  0.0873 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.558 on 70 degrees of freedom
## Multiple R-squared:  0.04119,    Adjusted R-squared:  0.0275
## F-statistic: 3.007 on 1 and 70 DF,  p-value: 0.08729
```

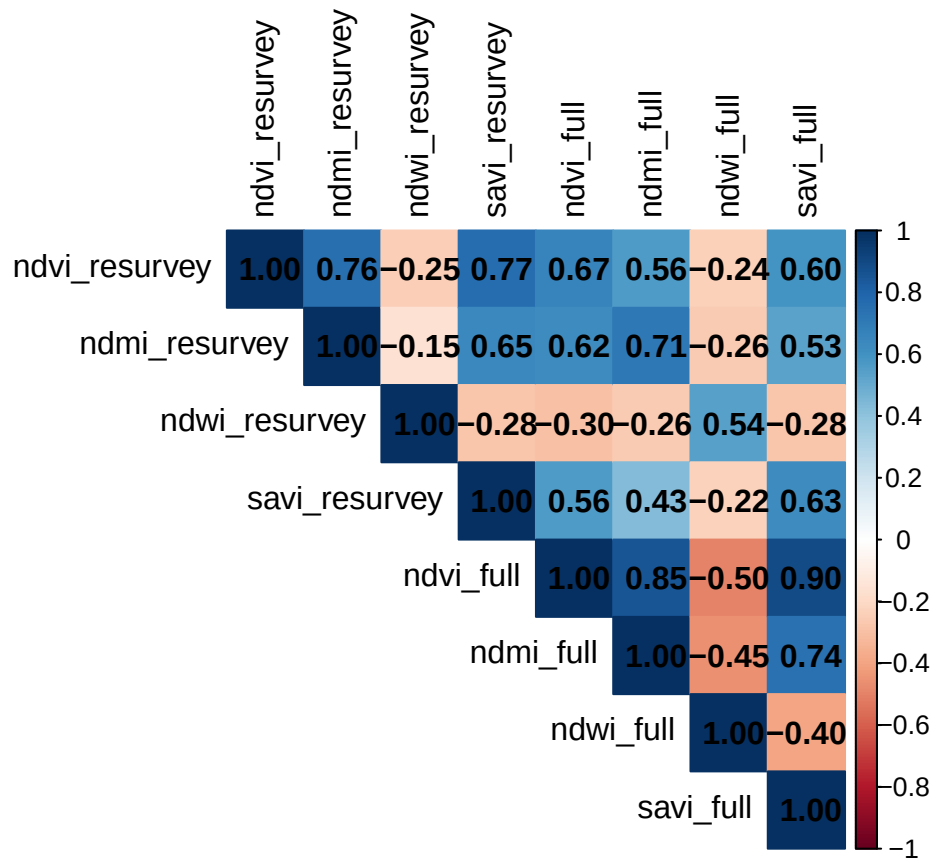
```
summary(mod_extent_yrs_ch_CON)
```

```
##
## Call:
## lm(formula = abs(delta_focal_pct) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == T) %>% filter(biogeo == "CON"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.156  -7.837  -3.436   3.145  56.195
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.61842     1.53589   6.914 9.13e-11 ***
## years_between.x -0.04854     0.13194  -0.368  0.713
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.14 on 170 degrees of freedom
## Multiple R-squared:  0.0007953,    Adjusted R-squared:  -0.005082
## F-statistic: 0.1353 on 1 and 170 DF,  p-value: 0.7134
```

Change in functional metrics

Correlations among functional metrics

```
RF_data %>%
  dplyr::select(matches("^ndvi|savi|ndmi|ndwi")) %>%
  cor(use = "pairwise.complete.obs") %>%
  corrplot(method = "color", type = "upper", addCoef.col = "black",
           number.cex = 1, tl.col = "black")
```



Plots with constant habitat

Change in functional metrics by habitat type

```
mod_ndvi_eunis1_noch <- lm(ndvi_resurvey ~ eunis1,
                           data = RF_data %>% filter(eunis_cambio == F))
mod_savi_eunis1_noch <- lm(savi_resurvey ~ eunis1,
                           data = RF_data %>% filter(eunis_cambio == F))
mod_ndmi_eunis1_noch <- lm(ndmi_resurvey ~ eunis1,
                           data = RF_data %>% filter(eunis_cambio == F))
mod_ndwi_eunis1_noch <- lm(ndwi_resurvey ~ eunis1,
                           data = RF_data %>% filter(eunis_cambio == F))
Anova(mod_ndvi_eunis1_noch)
```

```
## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.013943    3  824.95 < 2.2e-16 ***
## Residuals 0.060466 10733
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_eunis1_noch)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.010294    3  614.57 < 2.2e-16 ***
## Residuals 0.059926 10733
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndmi_eunis1_noch)
```

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.027300    3 1063.6 < 2.2e-16 ***
## Residuals 0.091834 10733
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_eunis1_noch)
```

```
## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.00863    3  277.91 < 2.2e-16 ***
## Residuals 0.11110 10733
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_ndvi_eunis1_noch, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1  emmean      SE    df lower.CL upper.CL
## Q      3.43e-03 1.01e-04 10733 3.24e-03 0.003634
## R      7.62e-05 2.96e-05 10733 1.82e-05 0.000134
## S      3.95e-03 9.14e-05 10733 3.77e-03 0.004132
## T      3.41e-04 4.27e-05 10733 2.57e-04 0.000424
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## Q - R      0.003358 0.000106 10733  31.768 <0.0001
## Q - S     -0.000518 0.000137 10733  -3.796  0.0009
## Q - T      0.003094 0.000110 10733  28.100 <0.0001
## R - S     -0.003877 0.000096 10733 -40.367 <0.0001
## R - T     -0.000265 0.000052 10733  -5.090 <0.0001
```

```
## S - T      0.003612 0.000101 10733  35.820 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_savi_eunis1_noch, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE      df lower.CL upper.CL
## Q          0.002341 1.01e-04 10733  0.002143  0.002539
## R          -0.000672 2.95e-05 10733 -0.000730 -0.000615
## S           0.002425 9.09e-05 10733  0.002247  0.002603
## T          -0.000739 4.25e-05 10733 -0.000822 -0.000656
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE      df t.ratio p.value
## Q - R      3.01e-03 1.05e-04 10733  28.634 <0.0001
## Q - S     -8.37e-05 1.36e-04 10733   -0.615  0.9272
## Q - T      3.08e-03 1.10e-04 10733  28.102 <0.0001
## R - S     -3.10e-03 9.56e-05 10733 -32.396 <0.0001
## R - T      6.67e-05 5.17e-05 10733   1.289  0.5700
## S - T      3.16e-03 1.00e-04 10733  31.516 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndmi_eunis1_noch, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE      df lower.CL upper.CL
## Q          0.002563 1.25e-04 10733  0.00232  0.002808
## R          -0.001041 3.65e-05 10733 -0.00111 -0.000969
## S           0.004976 1.13e-04 10733  0.00476  0.005197
## T           0.000233 5.26e-05 10733  0.00013  0.000336
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE      df t.ratio p.value
## Q - R      0.00360 0.000130 10733  27.662 <0.0001
## Q - S     -0.00241 0.000168 10733 -14.340 <0.0001
## Q - T      0.00233 0.000136 10733  17.174 <0.0001
## R - S     -0.00602 0.000118 10733 -50.839 <0.0001
## R - T     -0.00127 0.000064 10733 -19.889 <0.0001
## S - T      0.00474 0.000124 10733  38.168 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndwi_eunis1_noch, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE      df lower.CL upper.CL
```

```
## Q      -0.00414 1.38e-04 10733 -0.00441 -0.00387
## R      -0.00141 4.01e-05 10733 -0.00149 -0.00133
## S      -0.00321 1.24e-04 10733 -0.00346 -0.00297
## T      -0.00299 5.79e-05 10733 -0.00310 -0.00288
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## Q - R      -0.002734 1.43e-04 10733 -19.076 <0.0001
## Q - S      -0.000929 1.85e-04 10733  -5.017 <0.0001
## Q - T      -0.001153 1.49e-04 10733  -7.725 <0.0001
## R - S       0.001805 1.30e-04 10733  13.865 <0.0001
## R - T       0.001581 7.04e-05 10733  22.441 <0.0001
## S - T      -0.000224 1.37e-04 10733  -1.641  0.3555
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndvi_eunis1_noch)$r.squared
```

```
## [1] 0.1873776
```

```
summary(mod_savi_eunis1_noch)$r.squared
```

```
## [1] 0.1465969
```

```
summary(mod_ndmi_eunis1_noch)$r.squared
```

```
## [1] 0.2291544
```

```
summary(mod_ndwi_eunis1_noch)$r.squared
```

```
## [1] 0.07207886
```

Figure 2

```
fig2 <- ggplot(RF_data %>% filter(eunis_cambio == F) %>%
  pivot_longer(cols = c(ndvi_resurvey, savi_resurvey,
    ndmi_resurvey, ndwi_resurvey),
    names_to = "index", values_to = "value") %>%
  mutate(index = factor(index,
    levels = c("ndvi_resurvey",
      "savi_resurvey",
      "ndmi_resurvey",
      "ndwi_resurvey"),
    labels = c("NDVI", "SAVI", "NDMI",
      "NDWI")))),
```

```

aes(x = eunis1, y = value, fill = eunis1)) +
geom_point(aes(color = eunis1),
            position = position_jitter(width = 0.25), size = 0.5, alpha = 0.1) +
stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
             linewidth = 0.3, color = "black") +
stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
             color = "black", fill = "black") +
facet_wrap(~ index, ncol = 2, scales = "free") +
geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
labs(x = "EUNIS I Habitat type", y = "Sen's slope") +
theme_bw() +
theme(legend.position = "none")
fig2

```

```
## Warning: Removed 2068 rows containing non-finite outside the scale range
```

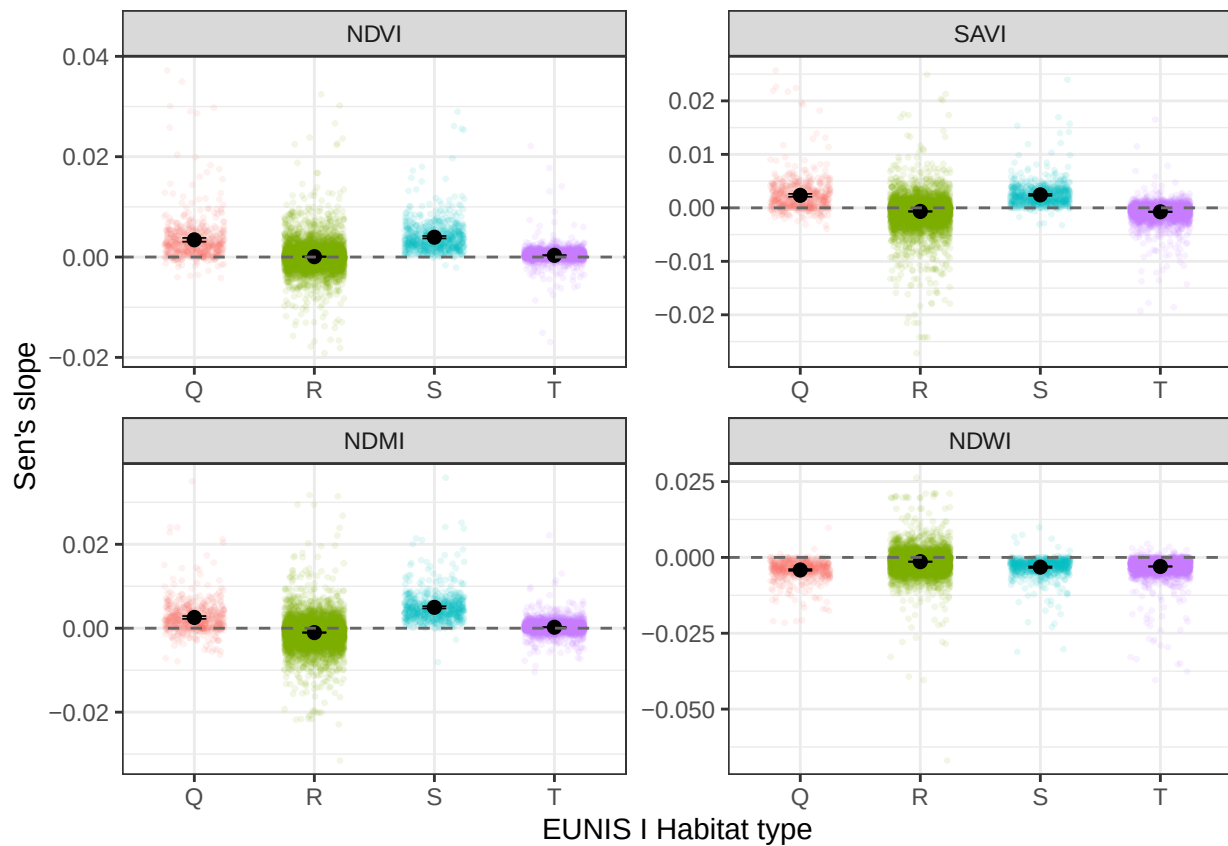
```
## ('stat_summary()').
```

```
## Removed 2068 rows containing non-finite outside the scale range
```

```
## ('stat_summary()').
```

```
## Warning: Removed 2068 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



```

ggsave(fig2, filename = "output/figures/fig2.jpeg",
        width = 24, height = 24, units = "cm", dpi = 300)

```

```
## Warning: Removed 2068 rows containing non-finite outside the scale range
## ('stat_summary()').

## Warning: Removed 2068 rows containing non-finite outside the scale range
## ('stat_summary()').

## Warning: Removed 2068 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Level II

```
# Prepare data
plot_data_eunis2 <- RF_data %>%
  filter(eunis_cambio == FALSE) %>%
  filter(eunis2 != "Change") %>%
  pivot_longer(cols = c(ndvi_resurvey, savi_resurvey, ndmi_resurvey,
                        ndwi_resurvey),
               names_to = "index", values_to = "value") %>%
  mutate(index = factor(index,
                        levels = c("ndvi_resurvey", "savi_resurvey",
                                   "ndmi_resurvey", "ndwi_resurvey"),
                        labels = c("NDVI", "SAVI", "NDMI", "NDWI")))

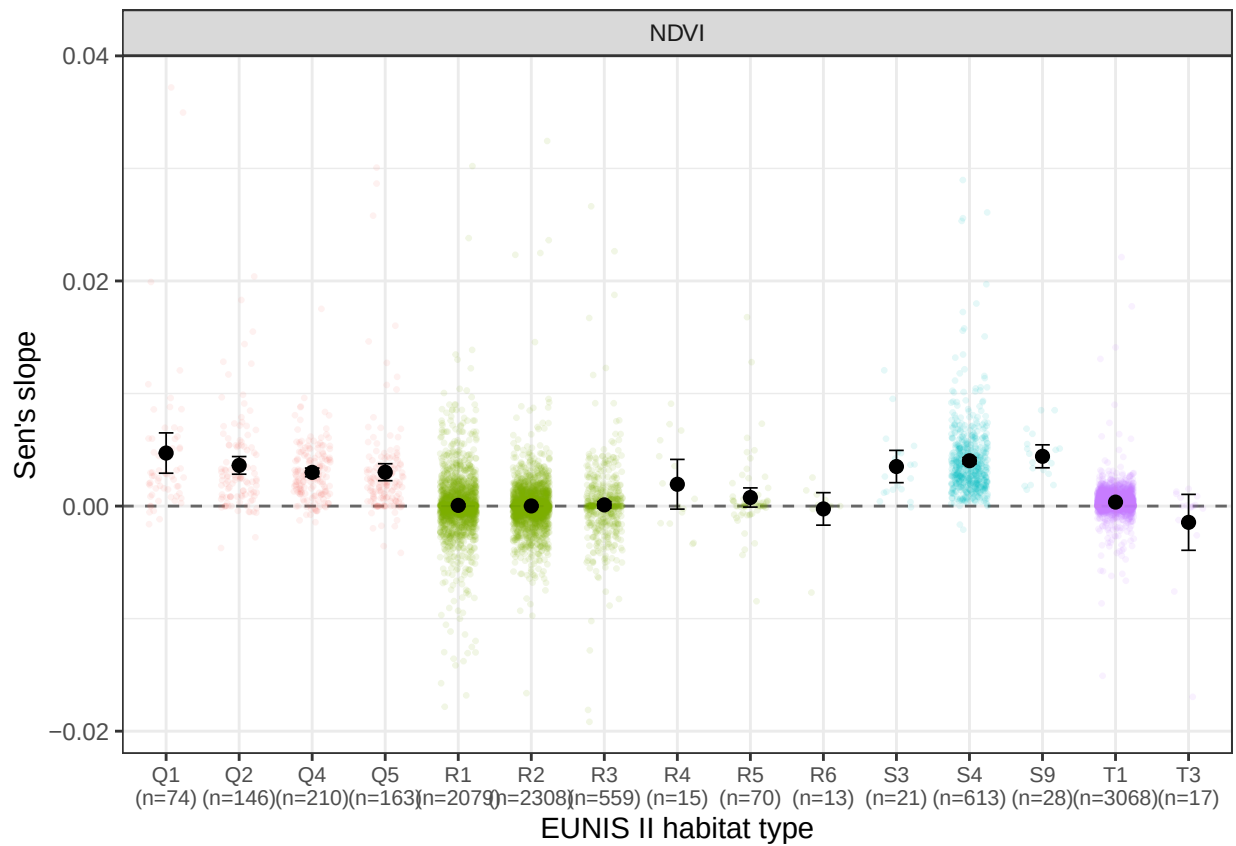
# Sample sizes per EUNIS II class
eunis2_counts <- plot_data_eunis2 %>%
  distinct(plot_id, eunis2) %>% # avoids counting same plot several times
  count(eunis2)

# Add labels containing habitat name + n
plot_data_eunis2 <- plot_data_eunis2 %>%
  left_join(eunis2_counts, by = "eunis2") %>%
  mutate(eunis2_label = paste0(eunis2, "\n(n=", n, ")"))

ggplot(plot_data_eunis2 %>%
  # Keep eunis2 with > 10 observations
  filter(n > 10) %>%
  filter(index == "NDVI"),
  aes(x = eunis2_label, y = value, fill = eunis1)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis1), position = position_jitter(width = 0.25),
             size = 0.5, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
              linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
              color = "black", fill = "black") +
  facet_wrap(~index, ncol = 1, scales = "free") +
  labs(x = "EUNIS II habitat type", y = "Sen's slope") +
  theme_bw() +
  theme(legend.position = "none",
        axis.text.x = element_text(size = 8, lineheight = 0.9))
```

```
## Warning: Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').

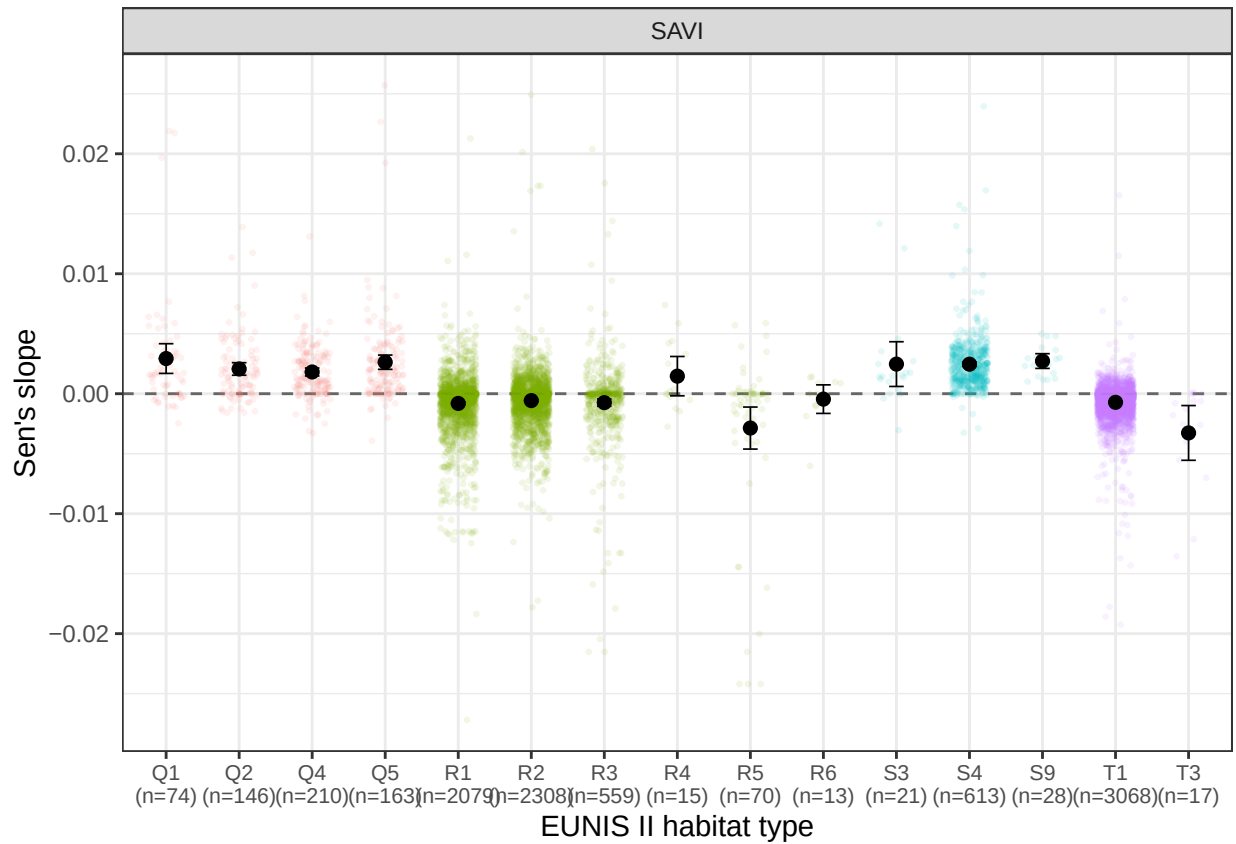
## Warning: Removed 442 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
ggplot(plot_data_eunis2 %>%
  # Keep eunis2 with > 10 observations
  filter(n > 10) %>%
  filter(index == "SAVI"),
  aes(x = eunis2_label, y = value, fill = eunis1)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis1), position = position_jitter(width = 0.25),
    size = 0.5, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  facet_wrap(~index, ncol = 1, scales = "free") +
  labs(x = "EUNIS II habitat type", y = "Sen's slope") +
  theme_bw() +
  theme(legend.position = "none",
    axis.text.x = element_text(size = 8, lineheight = 0.9))
```

```
## Warning: Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').

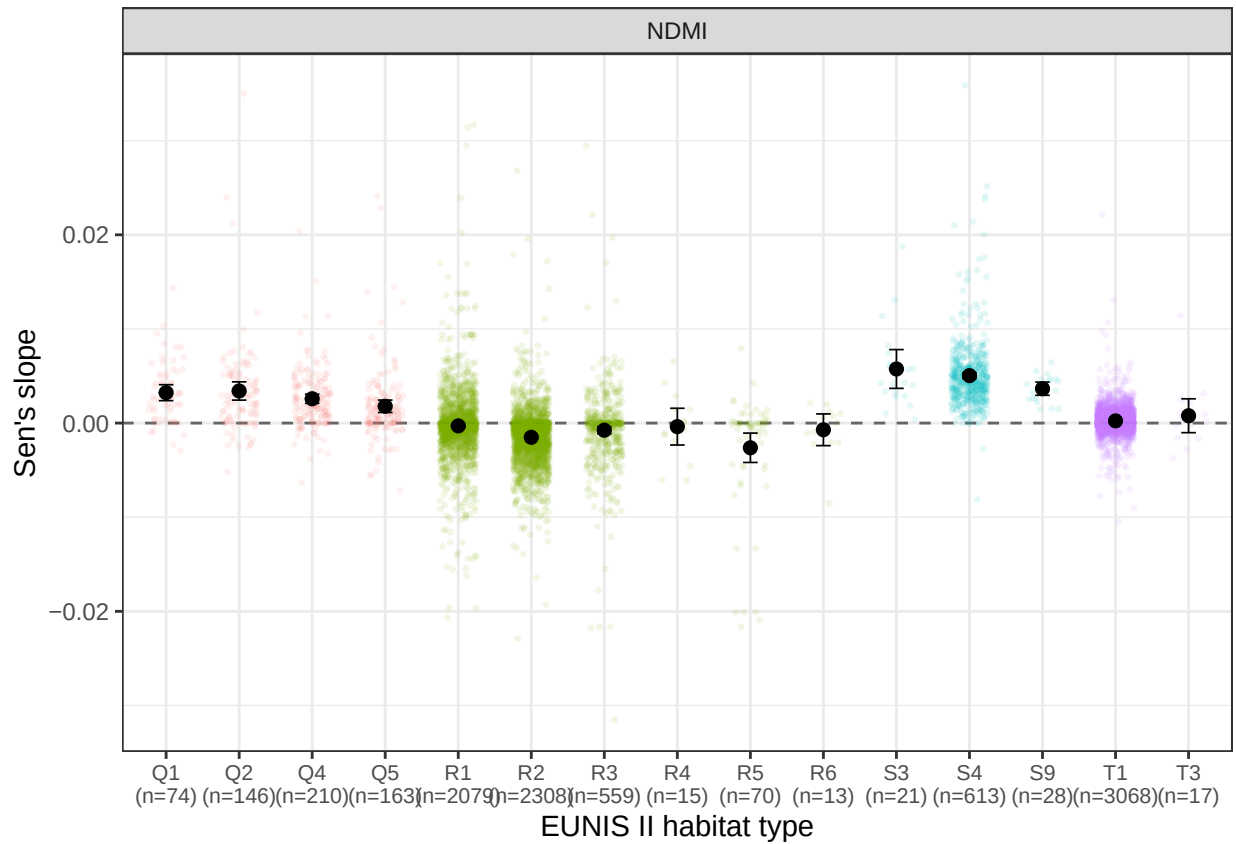
## Warning: Removed 442 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
ggplot(plot_data_eunis2 %>%
  # Keep eunis2 with > 10 observations
  filter(n > 10) %>%
  filter(index == "NDMI"),
  aes(x = eunis2_label, y = value, fill = eunis1)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis1), position = position_jitter(width = 0.25),
    size = 0.5, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  facet_wrap(~index, ncol = 1, scales = "free") +
  labs(x = "EUNIS II habitat type", y = "Sen's slope") +
  theme_bw() +
  theme(legend.position = "none",
    axis.text.x = element_text(size = 8, lineheight = 0.9))
```

```
## Warning: Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').

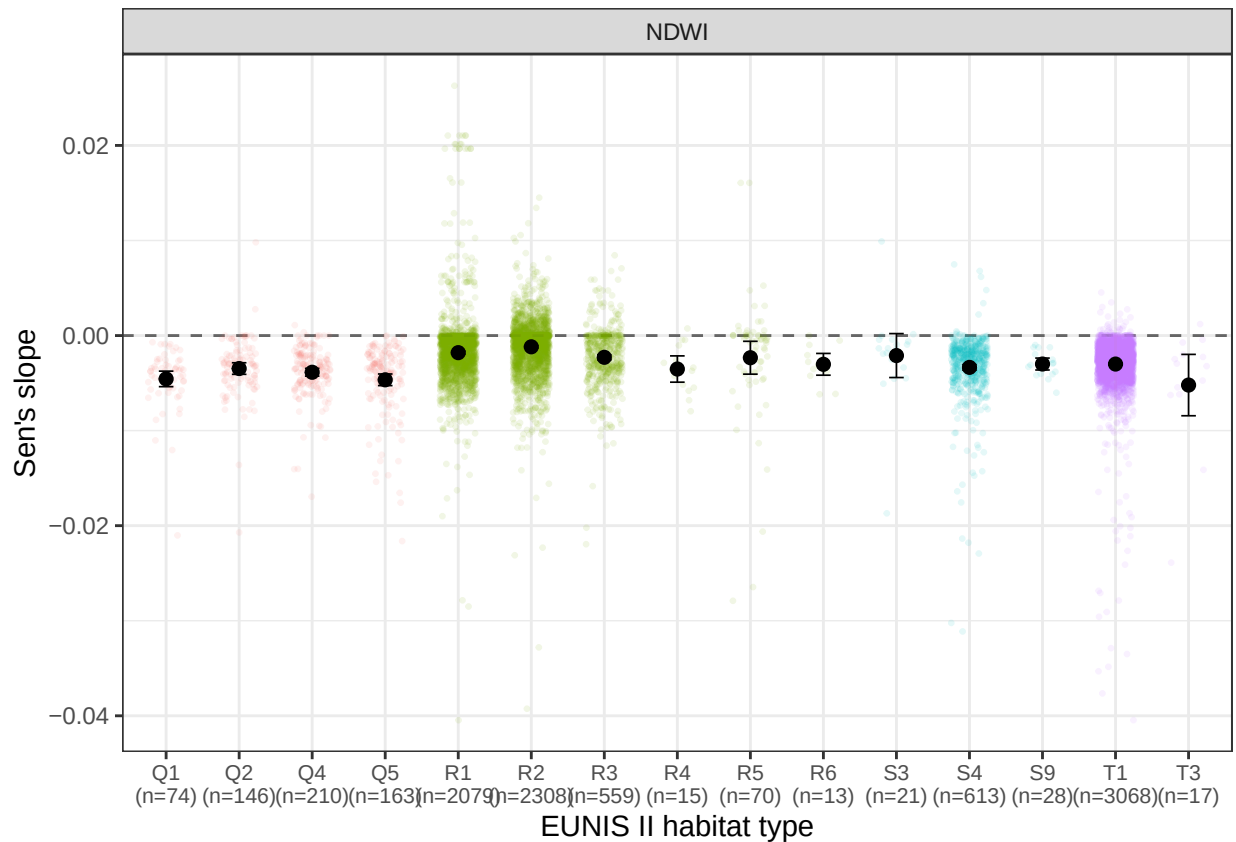
## Warning: Removed 442 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
ggplot(plot_data_eunis2 %>%
  # Keep eunis2 with > 10 observations
  filter(n > 10) %>%
  filter(index == "NDWI"),
  aes(x = eunis2_label, y = value, fill = eunis1)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis1), position = position_jitter(width = 0.25),
    size = 0.5, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  facet_wrap(~index, ncol = 1, scales = "free") +
  labs(x = "EUNIS II habitat type", y = "Sen's slope") +
  theme_bw() +
  theme(legend.position = "none",
    axis.text.x = element_text(size = 8, lineheight = 0.9))
```

```
## Warning: Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 442 rows containing non-finite outside the scale range
## ('stat_summary()').

## Warning: Removed 442 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Test with R

```
mod_ndvi_eunis2R_noch <- lm(ndvi_resurvey ~ eunis2,
  data = RF_data %>%
    filter(eunis_cambio == F & eunis1 == "R" &
      eunis2 != "Change"))
mod_savi_eunis2R_noch <- lm(savi_resurvey ~ eunis2,
  data = RF_data %>%
    filter(eunis_cambio == F & eunis1 == "R" &
      eunis2 != "Change"))
mod_ndmi_eunis2R_noch <- lm(ndmi_resurvey ~ eunis2,
  data = RF_data %>%
    filter(eunis_cambio == F & eunis1 == "R" &
      eunis2 != "Change"))
mod_ndwi_eunis2R_noch <- lm(ndwi_resurvey ~ eunis2,
```

```

data = RF_data %>%
  filter(eunis_cambio == F & eunis1 == "R" &
         eunis2 != "Change"))
Anova(mod_ndvi_eunis2R_noch)

```

```

## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis2    0.0000919    5  2.8948 0.01295 *
## Residuals 0.0309668 4878
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_savi_eunis2R_noch)
```

```

## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis2    0.000431    5 13.096 1.087e-12 ***
## Residuals 0.032112 4878
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndmi_eunis2R_noch)
```

```

## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis2    0.001761    5 30.917 < 2.2e-16 ***
## Residuals 0.055577 4878
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndwi_eunis2R_noch)
```

```

## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis2    0.000837    5 14.747 2.205e-14 ***
## Residuals 0.055388 4878
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
emmeans(mod_ndvi_eunis2R_noch, pairwise ~ eunis2)
```

```
## $emmeans
## eunis2      emmean      SE    df  lower.CL upper.CL
## R1      6.15e-05 5.65e-05 4878 -4.93e-05 0.000172
## R2      1.24e-05 5.29e-05 4878 -9.12e-05 0.000116
## R3      1.06e-04 1.09e-04 4878 -1.08e-04 0.000320
## R4      1.94e-03 6.51e-04 4878  6.61e-04 0.003212
## R5      7.62e-04 3.15e-04 4878  1.45e-04 0.001380
## R6     -2.52e-04 6.99e-04 4878 -1.62e-03 0.001118
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## R1 - R2  4.91e-05 7.74e-05 4878   0.634 0.9885
## R1 - R3 -4.43e-05 1.23e-04 4878  -0.361 0.9992
## R1 - R4 -1.88e-03 6.53e-04 4878  -2.872 0.0471
## R1 - R5 -7.01e-04 3.20e-04 4878  -2.190 0.2427
## R1 - R6  3.14e-04 7.01e-04 4878   0.447 0.9978
## R2 - R3 -9.34e-05 1.21e-04 4878  -0.771 0.9724
## R2 - R4 -1.92e-03 6.53e-04 4878  -2.948 0.0378
## R2 - R5 -7.50e-04 3.19e-04 4878  -2.347 0.1754
## R2 - R6  2.65e-04 7.01e-04 4878   0.378 0.9990
## R3 - R4 -1.83e-03 6.60e-04 4878  -2.776 0.0616
## R3 - R5 -6.56e-04 3.33e-04 4878  -1.969 0.3604
## R3 - R6  3.58e-04 7.07e-04 4878   0.506 0.9960
## R4 - R5  1.17e-03 7.23e-04 4878   1.625 0.5818
## R4 - R6  2.19e-03 9.55e-04 4878   2.293 0.1971
## R5 - R6  1.01e-03 7.66e-04 4878   1.323 0.7722
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
emmeans(mod_savi_eunis2R_noch, pairwise ~ eunis2)
```

```
## $emmeans
## eunis2      emmean      SE    df  lower.CL upper.CL
## R1     -0.000810 5.76e-05 4878 -0.000923 -0.000697
## R2     -0.000573 5.38e-05 4878 -0.000678 -0.000467
## R3     -0.000743 1.11e-04 4878 -0.000960 -0.000525
## R4      0.001465 6.62e-04 4878  0.000166  0.002763
## R5     -0.002861 3.21e-04 4878 -0.003490 -0.002233
## R6     -0.000451 7.12e-04 4878 -0.001846  0.000944
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## R1 - R2 -2.37e-04 7.88e-05 4878  -3.005 0.0319
## R1 - R3 -6.69e-05 1.25e-04 4878  -0.535 0.9947
## R1 - R4 -2.27e-03 6.65e-04 4878  -3.420 0.0083
## R1 - R5  2.05e-03 3.26e-04 4878   6.297 <0.0001
## R1 - R6 -3.59e-04 7.14e-04 4878  -0.503 0.9961
## R2 - R3  1.70e-04 1.23e-04 4878   1.377 0.7410
## R2 - R4 -2.04e-03 6.65e-04 4878  -3.065 0.0266
## R2 - R5  2.29e-03 3.25e-04 4878   7.037 <0.0001
```

```
## R2 - R6 -1.22e-04 7.14e-04 4878 -0.171 1.0000
## R3 - R4 -2.21e-03 6.72e-04 4878 -3.286 0.0131
## R3 - R5 2.12e-03 3.39e-04 4878 6.242 <0.0001
## R3 - R6 -2.92e-04 7.20e-04 4878 -0.406 0.9986
## R4 - R5 4.33e-03 7.36e-04 4878 5.877 <0.0001
## R4 - R6 1.92e-03 9.72e-04 4878 1.970 0.3599
## R5 - R6 -2.41e-03 7.81e-04 4878 -3.088 0.0247
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
emmeans(mod_ndmi_eunis2R_noch, pairwise ~ eunis2)
```

```
## $emmeans
## eunis2      emmean      SE    df lower.CL upper.CL
## R1      -0.000299 7.57e-05 4878 -0.000447 -0.000151
## R2      -0.001511 7.08e-05 4878 -0.001650 -0.001372
## R3      -0.000756 1.46e-04 4878 -0.001042 -0.000470
## R4      -0.000380 8.72e-04 4878 -0.002088 0.001329
## R5      -0.002625 4.22e-04 4878 -0.003452 -0.001797
## R6      -0.000711 9.36e-04 4878 -0.002546 0.001124
##
```

```
## Confidence level used: 0.95
```

```
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## R1 - R2 1.21e-03 0.000104 4878 11.687 <0.0001
## R1 - R3 4.57e-04 0.000165 4878 2.777 0.0613
## R1 - R4 8.05e-05 0.000875 4878 0.092 1.0000
## R1 - R5 2.33e-03 0.000429 4878 5.425 <0.0001
## R1 - R6 4.12e-04 0.000939 4878 0.439 0.9980
## R2 - R3 -7.55e-04 0.000162 4878 -4.650 <0.0001
## R2 - R4 -1.13e-03 0.000874 4878 -1.294 0.7886
## R2 - R5 1.11e-03 0.000428 4878 2.603 0.0966
## R2 - R6 -8.00e-04 0.000939 4878 -0.852 0.9576
## R3 - R4 -3.76e-04 0.000884 4878 -0.426 0.9982
## R3 - R5 1.87e-03 0.000446 4878 4.185 0.0004
## R3 - R6 -4.49e-05 0.000947 4878 -0.047 1.0000
## R4 - R5 2.24e-03 0.000968 4878 2.318 0.1867
## R4 - R6 3.31e-04 0.001280 4878 0.259 0.9998
## R5 - R6 -1.91e-03 0.001030 4878 -1.863 0.4250
##
```

```
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
emmeans(mod_ndwi_eunis2R_noch, pairwise ~ eunis2)
```

```
## $emmeans
## eunis2      emmean      SE    df lower.CL upper.CL
## R1      -0.00180 7.56e-05 4878 -0.00195 -0.00165
## R2      -0.00119 7.07e-05 4878 -0.00132 -0.00105
## R3      -0.00229 1.46e-04 4878 -0.00257 -0.00200
## R4      -0.00352 8.70e-04 4878 -0.00523 -0.00182
## R5      -0.00233 4.21e-04 4878 -0.00316 -0.00151
## R6      -0.00303 9.35e-04 4878 -0.00486 -0.00119
```

```
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## R1 - R2 -6.14e-04 0.000104 4878 -5.932 <0.0001
## R1 - R3 4.89e-04 0.000164 4878 2.977 0.0347
## R1 - R4 1.72e-03 0.000873 4878 1.973 0.3583
## R1 - R5 5.33e-04 0.000428 4878 1.246 0.8140
## R1 - R6 1.23e-03 0.000938 4878 1.308 0.7809
## R2 - R3 1.10e-03 0.000162 4878 6.806 <0.0001
## R2 - R4 2.34e-03 0.000873 4878 2.677 0.0801
## R2 - R5 1.15e-03 0.000427 4878 2.686 0.0781
## R2 - R6 1.84e-03 0.000937 4878 1.963 0.3637
## R3 - R4 1.23e-03 0.000882 4878 1.398 0.7281
## R3 - R5 4.43e-05 0.000446 4878 0.099 1.0000
## R3 - R6 7.37e-04 0.000946 4878 0.779 0.9710
## R4 - R5 -1.19e-03 0.000967 4878 -1.230 0.8220
## R4 - R6 -4.96e-04 0.001280 4878 -0.389 0.9989
## R5 - R6 6.93e-04 0.001030 4878 0.676 0.9846
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

Plots with habitat change

Change in functional metrics by habitat type

```
mod_ndvi_eunis1_ch <- lm(ndvi_resurvey ~ eunis_first,
  data = RF_data %>% filter(eunis_cambio == T))
mod_savi_eunis1_ch <- lm(savi_resurvey ~ eunis_first,
  data = RF_data %>% filter(eunis_cambio == T))
mod_ndmi_eunis1_ch <- lm(ndmi_resurvey ~ eunis_first,
  data = RF_data %>% filter(eunis_cambio == T))
mod_ndwi_eunis1_ch <- lm(ndwi_resurvey ~ eunis_first,
  data = RF_data %>% filter(eunis_cambio == T))
Anova(mod_ndvi_eunis1_ch)
```

```
## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
## Sum Sq Df F value Pr(>F)
## eunis_first 0.00009631 3 6.7494 0.000233 ***
## Residuals 0.00095126 200
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_eunis1_ch)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
```

```
##               Sum Sq Df F value    Pr(>F)
## eunis_first 0.00013897   3  11.905 3.313e-07 ***
## Residuals   0.00077821 200
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndmi_eunis1_ch)
```

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##               Sum Sq Df F value    Pr(>F)
## eunis_first 0.00007812   3  4.9372 0.002496 **
## Residuals   0.00105488 200
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_eunis1_ch)
```

```
## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##               Sum Sq Df F value    Pr(>F)
## eunis_first 0.00009939   3  5.8548 0.000749 ***
## Residuals   0.00113168 200
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_ndvi_eunis1_ch, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first emmean      SE df lower.CL upper.CL
## Q           0.002226 0.000263 200  0.001708  0.00274
## R           0.000948 0.000232 200  0.000490  0.00141
## S           0.002461 0.000354 200  0.001763  0.00316
## T           0.000899 0.000727 200 -0.000535  0.00233
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R      1.28e-03 0.000351 200   3.644  0.0019
## Q - S     -2.35e-04 0.000441 200  -0.533  0.9509
## Q - T      1.33e-03 0.000773 200   1.717  0.3175
## R - S     -1.51e-03 0.000423 200  -3.574  0.0025
## R - T      4.92e-05 0.000763 200   0.064  0.9999
## S - T      1.56e-03 0.000808 200   1.932  0.2179
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_savi_eunis1_ch, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first      emmean      SE df lower.CL upper.CL
## Q              0.001590 0.000237 200  0.001121  0.002058
## R              0.000201 0.000210 200 -0.000213  0.000616
## S              0.001513 0.000320 200  0.000882  0.002144
## T             -0.001447 0.000658 200 -0.002743 -0.000150
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R      1.39e-03 0.000317 200   4.377  0.0001
## Q - S      7.65e-05 0.000398 200   0.192  0.9975
## Q - T      3.04e-03 0.000699 200   4.343  0.0001
## R - S     -1.31e-03 0.000383 200  -3.426  0.0041
## R - T      1.65e-03 0.000690 200   2.387  0.0828
## S - T      2.96e-03 0.000731 200   4.048  0.0004
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndmi_eunis1_ch, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first      emmean      SE df lower.CL upper.CL
## Q              0.001800 0.000276 200  0.001255  0.00235
## R              0.000867 0.000245 200  0.000384  0.00135
## S              0.002466 0.000373 200  0.001731  0.00320
## T              0.001132 0.000766 200 -0.000377  0.00264
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R      0.000933 0.000369 200   2.526  0.0589
## Q - S     -0.000666 0.000464 200  -1.435  0.4790
## Q - T      0.000668 0.000814 200   0.821  0.8446
## R - S     -0.001599 0.000446 200  -3.586  0.0024
## R - T     -0.000265 0.000804 200  -0.330  0.9876
## S - T      0.001334 0.000851 200   1.567  0.4001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndwi_eunis1_ch, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first      emmean      SE df lower.CL upper.CL
## Q             -0.00362 0.000286 200 -0.00418 -0.00305
## R             -0.00207 0.000254 200 -0.00257 -0.00157
## S             -0.00243 0.000386 200 -0.00319 -0.00166
```

```
## T          -0.00331 0.000793 200 -0.00488 -0.00175
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R    -0.001549 0.000382 200  -4.050  0.0004
## Q - S    -0.001193 0.000481 200  -2.484  0.0656
## Q - T    -0.000307 0.000843 200  -0.364  0.9835
## R - S      0.000355 0.000462 200   0.770  0.8680
## R - T      0.001242 0.000832 200   1.492  0.4441
## S - T      0.000887 0.000882 200   1.006  0.7463
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndvi_eunis1_ch)$r.squared
```

```
## [1] 0.09193371
```

```
summary(mod_savi_eunis1_ch)$r.squared
```

```
## [1] 0.1515179
```

```
summary(mod_ndmi_eunis1_ch)$r.squared
```

```
## [1] 0.06895112
```

```
summary(mod_ndwi_eunis1_ch)$r.squared
```

```
## [1] 0.08073258
```

```
app3 <- ggplot(RF_data %>% filter(eunis_cambio == T) %>%
  pivot_longer(cols = c(ndvi_resurvey, savi_resurvey,
                        ndmi_resurvey, ndwi_resurvey),
               names_to = "index", values_to = "value") %>%
  mutate(index = factor(index,
                        levels = c("ndvi_resurvey",
                                   "savi_resurvey",
                                   "ndmi_resurvey",
                                   "ndwi_resurvey"),
                        labels = c("NDVI", "SAVI", "NDMI",
                                   "NDWI"))),
  aes(x = eunis_first, y = value, fill = eunis_first)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis_first),
             position = position_jitter(width = 0.25), size = 2, alpha = 0.2) +
```

```

stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
             linewidth = 0.3, color = "black") +
stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
             color = "black", fill = "black") +
geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
facet_wrap(~ index, nrow = 2, scales = "free") +
labs(x = "Original EUNIS I Habitat type", y = "Sen's slope") +
theme_bw() +
theme(legend.position = "none")
app3

```

Appendix 3

```

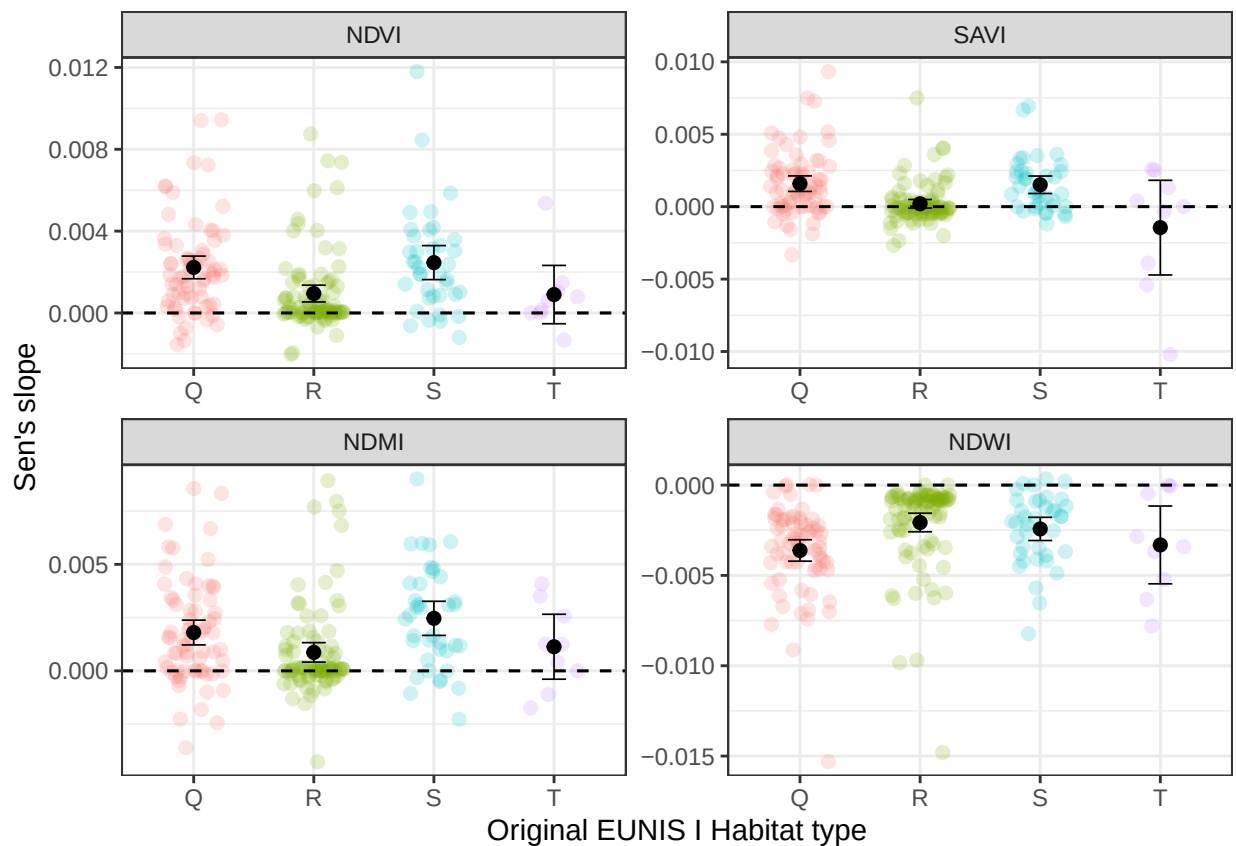
## Warning: Removed 244 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 244 rows containing non-finite outside the scale range
## ('stat_summary()').

```

```

## Warning: Removed 244 rows containing missing values or values outside the scale range
## ('geom_point()').

```



```

ggsave(app3, filename = "output/figures/app3.jpeg",
        width = 24, height = 24, units = "cm", dpi = 300)

```

```
## Warning: Removed 244 rows containing non-finite outside the scale range
## ('stat_summary()').

## Warning: Removed 244 rows containing non-finite outside the scale range
## ('stat_summary()').

## Warning: Removed 244 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Change in functional metrics by transition type

```
ggplot(RF_data %>% filter(eunis_cambio == TRUE) %>%
  left_join(RF_data %>% filter(eunis_cambio == TRUE) %>%
    count(transition),
    by = "transition") %>%
  mutate(transition_label = paste0(transition, "\n(n = ", n, ")")) %>%
  pivot_longer(cols = c(ndvi_resurvey, savi_resurvey, ndmi_resurvey,
    ndwi_resurvey),
    names_to = "index", values_to = "value") %>%
  mutate(index = factor(index,
    levels = c("ndvi_resurvey", "savi_resurvey",
      "ndmi_resurvey", "ndwi_resurvey"),
    labels = c("NDVI", "SAVI", "NDMI", "NDWI"))),
  aes(x = transition_label, y = value, fill = eunis_first)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(aes(color = eunis_first), position = position_jitter(width = 0.25),
    size = 2, alpha = 0.2) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  facet_wrap(~index, nrow = 2, scales = "free") +
  labs(x = "Transition", y = "Sen's slope") +
  theme_bw() + theme(axis.text.x = element_text(size = 8, lineheight = 0.9)) +
  theme(legend.position = "none")
```

```
## Warning: Removed 244 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 244 rows containing non-finite outside the scale range
## ('stat_summary()').

## Warning: Removed 244 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```

                                filter(biogeo == "ALP"))
mod_ndvi_eunis1_noch_ATL <- lm(ndvi_resurvey ~ eunis1,
                                data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "ATL"))
mod_ndvi_eunis1_noch_CON <- lm(ndvi_resurvey ~ eunis1,
                                data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "CON"))
# Not possible in MED because all are R
mod_ndvi_eunis1_noch_PAN <- lm(ndvi_resurvey ~ eunis1,
                                data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "PAN"))
Anova(mod_ndvi_eunis1_noch_ALP)

```

NDVI

```

## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq  Df F value Pr(>F)
## eunis1    0.00001613   2  1.4409 0.2379
## Residuals 0.00235050 420

```

```
Anova(mod_ndvi_eunis1_noch_ATL)
```

```

## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq  Df F value    Pr(>F)
## eunis1    0.0049296   3 189.19 < 2.2e-16 ***
## Residuals 0.0155038 1785
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndvi_eunis1_noch_CON)
```

```

## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq  Df F value    Pr(>F)
## eunis1    0.007019   3 517.28 < 2.2e-16 ***
## Residuals 0.036531 8077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndvi_eunis1_noch_PAN)
```

```

## Anova Table (Type II tests)
##

```

```
## Response: ndvi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis1    7.7700e-07 1  1.4139 0.2353
## Residuals 1.7645e-04 321
```

```
emmeans(mod_ndvi_eunis1_noch_ATL, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE    df lower.CL upper.CL
## Q        0.003949 2.51e-04 1785  3.46e-03 0.004441
## R        0.000503 9.43e-05 1785  3.18e-04 0.000688
## S        0.004023 1.38e-04 1785  3.75e-03 0.004294
## T        0.000382 1.99e-04 1785 -8.68e-06 0.000772
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## Q - R      3.45e-03 0.000268 1785  12.859 <0.0001
## Q - S     -7.37e-05 0.000286 1785   -0.257  0.9940
## Q - T      3.57e-03 0.000320 1785  11.138 <0.0001
## R - S     -3.52e-03 0.000167 1785 -21.045 <0.0001
## R - T      1.21e-04 0.000220 1785   0.550  0.9467
## S - T      3.64e-03 0.000242 1785  15.023 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndvi_eunis1_noch_CON, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE    df lower.CL upper.CL
## Q        3.29e-03 1.06e-04 8077  0.003084 3.50e-03
## R       -9.07e-05 3.11e-05 8077 -0.000152 -2.98e-05
## S        3.81e-03 1.43e-04 8077  0.003527 4.09e-03
## T        3.57e-04 4.04e-05 8077  0.000278 4.36e-04
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## Q - R      0.003383 0.000111 8077  30.604 <0.0001
## Q - S     -0.000516 0.000178 8077   -2.893  0.0200
## Q - T      0.002935 0.000114 8077  25.858 <0.0001
## R - S     -0.003898 0.000147 8077 -26.573 <0.0001
## R - T     -0.000448 0.000051 8077   -8.782 <0.0001
## S - T      0.003451 0.000149 8077  23.166 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndvi_eunis1_noch_ALP)$r.squared
```

```
## [1] 0.006814515
```

```
summary(mod_ndvi_eunis1_noch_ATL)$r.squared
```

```
## [1] 0.2412523
```

```
summary(mod_ndvi_eunis1_noch_CON)$r.squared
```

```
## [1] 0.1611654
```

```
summary(mod_ndvi_eunis1_noch_PAN)$r.squared
```

```
## [1] 0.004385401
```

```
mod_savi_eunis1_noch_ALP <- lm(savi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "ALP"))
mod_savi_eunis1_noch_ATL <- lm(savi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "ATL"))
mod_savi_eunis1_noch_CON <- lm(savi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "CON"))
# Not possible in MED because all are R
mod_savi_eunis1_noch_PAN <- lm(savi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "PAN"))
Anova(mod_savi_eunis1_noch_ALP)
```

SAVI

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq Df F value  Pr(>F)
## eunis1    0.0000324  2  2.7995 0.06197 .
## Residuals 0.0024301 420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_eunis1_noch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.003791    3  119.85 < 2.2e-16 ***
## Residuals 0.018821 1785
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_eunis1_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.005104    3  406.35 < 2.2e-16 ***
## Residuals 0.033814 8077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_eunis1_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.00011204    1  169.98 < 2.2e-16 ***
## Residuals 0.00021158 321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_savi_eunis1_noch_ATL, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE    df lower.CL upper.CL
## Q      0.002864 0.000276 1785  0.002322  3.41e-03
## R     -0.000568 0.000104 1785 -0.000772 -3.64e-04
## S      0.002451 0.000152 1785  0.002152  2.75e-03
## T     -0.000391 0.000219 1785 -0.000821  3.95e-05
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## Q - R      0.003432 0.000295 1785  11.624 <0.0001
## Q - S      0.000414 0.000316 1785   1.311  0.5562
## Q - T      0.003255 0.000353 1785   9.224 <0.0001
## R - S     -0.003019 0.000184 1785 -16.380 <0.0001
## R - T     -0.000177 0.000243 1785  -0.730  0.8850
```

```
## S - T      0.002842 0.000267 1785  10.640 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_savi_eunis1_noch_CON, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE   df lower.CL upper.CL
## Q         0.002191 1.02e-04 8077  0.001991  0.002391
## R        -0.000717 2.99e-05 8077 -0.000776 -0.000659
## S         0.002371 1.38e-04 8077  0.002101  0.002642
## T        -0.000739 3.89e-05 8077 -0.000815 -0.000663
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE   df t.ratio p.value
## Q - R      2.91e-03 0.000106 8077  27.348 <0.0001
## Q - S     -1.81e-04 0.000172 8077  -1.053  0.7184
## Q - T      2.93e-03 0.000109 8077  26.827 <0.0001
## R - S     -3.09e-03 0.000141 8077 -21.882 <0.0001
## R - T      2.15e-05 0.000049 8077   0.439  0.9718
## S - T      3.11e-03 0.000143 8077  21.701 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_savi_eunis1_noch_PAN, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE   df lower.CL upper.CL
## R        -0.000335 5.23e-05 321 -0.000438 -0.000232
## T        -0.001688 8.97e-05 321 -0.001864 -0.001512
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE   df t.ratio p.value
## R - T      0.00135 0.000104 321  13.038 <0.0001
```

Proportion of variance explained:

```
summary(mod_savi_eunis1_noch_ALP)$r.squared
```

```
## [1] 0.01315568
```

```
summary(mod_savi_eunis1_noch_ATL)$r.squared
```

```
## [1] 0.1676527
```

```
summary(mod_savi_eunis1_noch_CON)$r.squared
```

```
## [1] 0.1311366
```

```
summary(mod_savi_eunis1_noch_PAN)$r.squared
```

```
## [1] 0.346211
```

```
mod_ndmi_eunis1_noch_ALP <- lm(ndmi_resurvey ~ eunis1,
                               data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ALP"))
mod_ndmi_eunis1_noch_ATL <- lm(ndmi_resurvey ~ eunis1,
                               data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ATL"))
mod_ndmi_eunis1_noch_CON <- lm(ndmi_resurvey ~ eunis1,
                               data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "CON"))
# Not possible in MED because all are R
mod_ndmi_eunis1_noch_PAN <- lm(ndmi_resurvey ~ eunis1,
                               data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "PAN"))
Anova(mod_ndmi_eunis1_noch_ALP)
```

NDMI

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq  Df F value Pr(>F)
## eunis1    0.00000801  2  0.6064 0.5458
## Residuals 0.00277343 420
```

```
Anova(mod_ndmi_eunis1_noch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq  Df F value    Pr(>F)
## eunis1    0.0090531   3 231.98 < 2.2e-16 ***
## Residuals 0.0232205 1785
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndmi_eunis1_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq   Df F value    Pr(>F)
## eunis1    0.013890    3  683.82 < 2.2e-16 ***
## Residuals 0.054688 8077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndmi_eunis1_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq   Df F value Pr(>F)
## eunis1    0.00000030    1  0.2859 0.5933
## Residuals 0.00034082 321
```

```
emmeans(mod_ndmi_eunis1_noch_ATL, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE    df lower.CL upper.CL
## Q         4.02e-03 0.000307 1785  0.003419 0.004623
## R        -3.78e-05 0.000115 1785 -0.000264 0.000189
## S         5.09e-03 0.000169 1785  0.004761 0.005424
## T         9.55e-04 0.000244 1785  0.000477 0.001433
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE    df t.ratio p.value
## Q - R      0.004059 0.000328 1785  12.375 <0.0001
## Q - S     -0.001072 0.000351 1785   -3.058  0.0121
## Q - T      0.003066 0.000392 1785    7.822 <0.0001
## R - S     -0.005131 0.000205 1785  -25.063 <0.0001
## R - T     -0.000992 0.000270 1785   -3.680  0.0014
## S - T      0.004138 0.000297 1785   13.950 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndmi_eunis1_noch_CON, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1      emmean      SE    df lower.CL upper.CL
## Q         0.002090 1.30e-04 8077  0.001836 0.002345
## R        -0.001399 3.80e-05 8077 -0.001474 -0.001325
## S         0.004736 1.75e-04 8077  0.004392 0.005080
## T         0.000204 4.94e-05 8077  0.000107 0.000301
##
```

```
## Confidence level used: 0.95
##
## $contrasts
##   contrast estimate      SE    df t.ratio p.value
##   Q - R      0.00349 1.35e-04 8077   25.804 <0.0001
##   Q - S     -0.00265 2.18e-04 8077  -12.123 <0.0001
##   Q - T      0.00189 1.39e-04 8077   13.584 <0.0001
##   R - S     -0.00613 1.80e-04 8077  -34.178 <0.0001
##   R - T     -0.00160 6.24e-05 8077  -25.707 <0.0001
##   S - T      0.00453 1.82e-04 8077   24.865 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndmi_eunis1_noch_ALP)$r.squared
```

```
## [1] 0.002879475
```

```
summary(mod_ndmi_eunis1_noch_ATL)$r.squared
```

```
## [1] 0.2805108
```

```
summary(mod_ndmi_eunis1_noch_CON)$r.squared
```

```
## [1] 0.2025451
```

```
summary(mod_ndmi_eunis1_noch_PAN)$r.squared
```

```
## [1] 0.0008897625
```

```
mod_ndwi_eunis1_noch_ALP <- lm(ndwi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "ALP"))
mod_ndwi_eunis1_noch_ATL <- lm(ndwi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "ATL"))
mod_ndwi_eunis1_noch_CON <- lm(ndwi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "CON"))
# Not possible in MED because all are R
mod_ndwi_eunis1_noch_PAN <- lm(ndwi_resurvey ~ eunis1,
                              data = RF_data %>%
                                filter(eunis_cambio == F) %>%
                                filter(biogeo == "PAN"))
Anova(mod_ndwi_eunis1_noch_ALP)
```

NDWI

```
## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis1    0.0000308  2  1.0778 0.3413
## Residuals 0.0060091 420
```

```
Anova(mod_ndwi_eunis1_noch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis1    0.0026076  3 85.748 < 2.2e-16 ***
## Residuals 0.0180939 1785
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_eunis1_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis1    0.004856  3 180.69 < 2.2e-16 ***
## Residuals 0.072350 8077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_eunis1_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis1    6.0807e-05  1 159.35 < 2.2e-16 ***
## Residuals 1.2249e-04 321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_ndwi_eunis1_noch_ATL, pairwise ~ eunis1)
```

```
## $emmeans
## eunis1    emmean      SE   df lower.CL upper.CL
## Q      -0.003555 0.000271 1785 -0.00409 -0.003024
## R      -0.000905 0.000102 1785 -0.00111 -0.000706
## S      -0.003303 0.000149 1785 -0.00360 -0.003010
## T      -0.003229 0.000215 1785 -0.00365 -0.002807
##
```

```
## Confidence level used: 0.95
##
## $contrasts
##   contrast   estimate      SE    df t.ratio p.value
##   Q - R    -2.65e-03 0.000290 1785   -9.152 <0.0001
##   Q - S    -2.52e-04 0.000309 1785   -0.815  0.8472
##   Q - T    -3.26e-04 0.000346 1785   -0.941  0.7825
##   R - S     2.40e-03 0.000181 1785   13.267 <0.0001
##   R - T     2.32e-03 0.000238 1785    9.763 <0.0001
##   S - T    -7.35e-05 0.000262 1785   -0.281  0.9923
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndwi_eunis1_noch_CON, pairwise ~ eunis1)
```

```
## $emmeans
##   eunis1   emmean      SE    df lower.CL upper.CL
##   Q      -0.00439 1.49e-04 8077 -0.00468 -0.00410
##   R      -0.00173 4.37e-05 8077 -0.00182 -0.00165
##   S      -0.00303 2.02e-04 8077 -0.00343 -0.00264
##   T      -0.00304 5.69e-05 8077 -0.00315 -0.00293
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast   estimate      SE    df t.ratio p.value
##   Q - R    -2.65e-03 1.56e-04 8077 -17.068 <0.0001
##   Q - S    -1.36e-03 2.51e-04 8077  -5.413 <0.0001
##   Q - T    -1.35e-03 1.60e-04 8077  -8.470 <0.0001
##   R - S     1.30e-03 2.06e-04 8077   6.278 <0.0001
##   R - T     1.30e-03 7.17e-05 8077  18.154 <0.0001
##   S - T     5.81e-06 2.10e-04 8077   0.028  1.0000
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```
emmeans(mod_ndwi_eunis1_noch_PAN, pairwise ~ eunis1)
```

```
## $emmeans
##   eunis1   emmean      SE    df lower.CL upper.CL
##   R      -0.000312 3.98e-05 321 -0.00039 -0.000234
##   T      -0.001309 6.82e-05 321 -0.00144 -0.001175
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast   estimate      SE    df t.ratio p.value
##   R - T     0.000997 7.9e-05 321  12.624 <0.0001
```

Proportion of variance explained:

```
summary(mod_ndwi_eunis1_noch_ALP)$r.squared
```

```
## [1] 0.005106045
```

```
summary(mod_ndwi_eunis1_noch_ATL)$r.squared
```

```
## [1] 0.1259612
```

```
summary(mod_ndwi_eunis1_noch_CON)$r.squared
```

```
## [1] 0.06289068
```

```
summary(mod_ndwi_eunis1_noch_PAN)$r.squared
```

```
## [1] 0.3317431
```

Change in functional metrics by number of years between first and last survey

```
mod_ndvi_yrs_noch_ALP <- lm(abs(ndvi_resurvey) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeoe == "ALP"))  
mod_ndvi_yrs_noch_ATL <- lm(abs(ndvi_resurvey) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeoe == "ATL"))  
mod_ndvi_yrs_noch_CON <- lm(abs(ndvi_resurvey) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeoe == "CON"))  
mod_ndvi_yrs_noch_MED <- lm(abs(ndvi_resurvey) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeoe == "MED"))  
mod_ndvi_yrs_noch_PAN <- lm(abs(ndvi_resurvey) ~ years_between.x,  
                             data = RF_data %>%  
                             filter(eunis_cambio == F) %>%  
                             filter(biogeoe == "PAN"))  
Anova(mod_ndvi_yrs_noch_ALP)
```

NDVI

```
## Anova Table (Type II tests)  
##  
## Response: abs(ndvi_resurvey)  
##           Sum Sq Df F value    Pr(>F)  
## years_between.x 0.00025679  1    65.6 6.026e-15 ***  
## Residuals      0.00164802 421  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndvi_yrs_noch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndvi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 0.0032264    1  436.43 < 2.2e-16 ***
## Residuals      0.0132110 1787
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndvi_yrs_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndvi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 0.0022737    1  605.08 < 2.2e-16 ***
## Residuals      0.0303582 8079
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndvi_yrs_noch_MED)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndvi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 0.00014182    1   9.6813 0.002331 **
## Residuals      0.00174324 119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndvi_yrs_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndvi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 2.8222e-05    1  68.886 2.933e-15 ***
## Residuals      1.3151e-04 321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(mod_ndvi_yrs_noch_ALP)
```

```
##
## Call:
## lm(formula = abs(ndvi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ALP"))
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0023071 -0.0013255 -0.0004959  0.0005774  0.0121003
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.546e-03  1.470e-04  17.322 < 2e-16 ***
## years_between.x -7.969e-05  9.839e-06  -8.099 6.03e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001979 on 421 degrees of freedom
## (18 observations deleted due to missingness)
## Multiple R-squared:  0.1348, Adjusted R-squared:  0.1328
## F-statistic: 65.6 on 1 and 421 DF, p-value: 6.026e-15
```

```
summary(mod_ndvi_yrs_noch_ATL)
```

```
##
## Call:
## lm(formula = abs(ndvi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ATL"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0040998 -0.0017387 -0.0004133  0.0009462  0.0261092
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.357e-03  1.206e-04  36.13 <2e-16 ***
## years_between.x -2.567e-04  1.229e-05  -20.89 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002719 on 1787 degrees of freedom
## (134 observations deleted due to missingness)
## Multiple R-squared:  0.1963, Adjusted R-squared:  0.1958
## F-statistic: 436.4 on 1 and 1787 DF, p-value: < 2.2e-16
```

```
summary(mod_ndvi_yrs_noch_CON)
```

```
##
## Call:
## lm(formula = abs(ndvi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "CON"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.001923 -0.000944 -0.000454  0.000391  0.035274
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)      2.027e-03  3.960e-05   51.2   <2e-16 ***
## years_between.x -1.039e-04  4.225e-06  -24.6   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001938 on 8079 degrees of freedom
## (351 observations deleted due to missingness)
## Multiple R-squared:  0.06968,    Adjusted R-squared:  0.06956
## F-statistic: 605.1 on 1 and 8079 DF,  p-value: < 2.2e-16
```

```
summary(mod_ndvi_yrs_noch_MED)
```

```
##
## Call:
## lm(formula = abs(ndvi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "MED"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0053759 -0.0029065 -0.0002473  0.0022156  0.0103643
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.0059433   0.0004724   12.582 < 2e-16 ***
## years_between.x -0.0005674   0.0001823   -3.111  0.00233 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003827 on 119 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.07523,    Adjusted R-squared:  0.06746
## F-statistic: 9.681 on 1 and 119 DF,  p-value: 0.002331
```

```
summary(mod_ndvi_yrs_noch_PAN)
```

```
##
## Call:
## lm(formula = abs(ndvi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "PAN"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0008061 -0.0001767 -0.0001210 -0.0000765  0.0048757
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.226e-03  1.138e-04   10.77 < 2e-16 ***
## years_between.x -6.993e-05  8.426e-06   -8.30 2.93e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0006401 on 321 degrees of freedom
## (13 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.1767, Adjusted R-squared:  0.1741
## F-statistic: 68.89 on 1 and 321 DF,  p-value: 2.933e-15
```

```
mod_savi_yrs_noch_ALP <- lm(abs(savi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ALP"))
mod_savi_yrs_noch_ATL <- lm(abs(savi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ATL"))
mod_savi_yrs_noch_CON <- lm(abs(savi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "CON"))
mod_savi_yrs_noch_MED <- lm(abs(savi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "MED"))
mod_savi_yrs_noch_PAN <- lm(abs(savi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "PAN"))

Anova(mod_savi_yrs_noch_ALP)
```

SAVI

```
## Anova Table (Type II tests)
##
## Response: abs(savi_resurvey)
##              Sum Sq  Df F value    Pr(>F)
## years_between.x 0.00021738    1  70.165 8.208e-16 ***
## Residuals      0.00130432 421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_yrs_noch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: abs(savi_resurvey)
##              Sum Sq  Df F value    Pr(>F)
## years_between.x 0.0030357    1 402.19 < 2.2e-16 ***
## Residuals      0.0134881 1787
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_yrs_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: abs(savi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 0.002408    1  818.11 < 2.2e-16 ***
## Residuals      0.023779 8079
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_yrs_noch_MED)
```

```
## Anova Table (Type II tests)
##
## Response: abs(savi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 0.00024149    1  24.115 2.921e-06 ***
## Residuals      0.00119170 119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_savi_yrs_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: abs(savi_resurvey)
##           Sum Sq   Df F value    Pr(>F)
## years_between.x 0.00012547    1  205.49 < 2.2e-16 ***
## Residuals      0.00019601 321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(mod_savi_yrs_noch_ALP)
```

```
##
## Call:
## lm(formula = abs(savi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ALP"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0023677 -0.0012343 -0.0002833  0.0005299  0.0149602
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.441e-03  1.308e-04  18.667 < 2e-16 ***
## years_between.x -7.332e-05  8.753e-06  -8.376 8.21e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00176 on 421 degrees of freedom
## (18 observations deleted due to missingness)
## Multiple R-squared:  0.1429, Adjusted R-squared:  0.1408
## F-statistic: 70.16 on 1 and 421 DF, p-value: 8.208e-16
```

```
summary(mod_savi_yrs_noch_ATL)
```

```
##
## Call:
## lm(formula = abs(savi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ATL"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0037256 -0.0014509 -0.0003709  0.0007595  0.0204741
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.975e-03  1.218e-04   32.63  <2e-16 ***
## years_between.x -2.490e-04  1.241e-05  -20.05  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002747 on 1787 degrees of freedom
## (134 observations deleted due to missingness)
## Multiple R-squared:  0.1837, Adjusted R-squared:  0.1833
## F-statistic: 402.2 on 1 and 1787 DF,  p-value: < 2.2e-16
```

```
summary(mod_savi_yrs_noch_CON)
```

```
##
## Call:
## lm(formula = abs(savi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "CON"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0020833 -0.0009606 -0.0003600  0.0004925  0.0251027
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.190e-03  3.504e-05   62.5  <2e-16 ***
## years_between.x -1.070e-04  3.739e-06  -28.6  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001716 on 8079 degrees of freedom
## (351 observations deleted due to missingness)
## Multiple R-squared:  0.09195, Adjusted R-squared:  0.09184
## F-statistic: 818.1 on 1 and 8079 DF,  p-value: < 2.2e-16
```

```
summary(mod_savi_yrs_noch_MED)
```

```
##
## Call:
## lm(formula = abs(savi_resurvey) ~ years_between.x, data = RF_data %>%
```

```
## filter(eunis_cambio == F) %>% filter(biogeo == "MED"))
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.0055512 -0.0027695 -0.0003973  0.0017035  0.0069543
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.0062916  0.0003906  16.109 < 2e-16 ***
## years_between.x -0.0007404  0.0001508  -4.911 2.92e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003165 on 119 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.1685, Adjusted R-squared:  0.1615
## F-statistic: 24.11 on 1 and 119 DF, p-value: 2.921e-06
```

```
summary(mod_savi_yrs_noch_PAN)
```

```
##
## Call:
## lm(formula = abs(savi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "PAN"))
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.0022792 -0.0003457 -0.0003182  0.0002029  0.0036041
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.574e-03  1.389e-04  18.53 <2e-16 ***
## years_between.x -1.475e-04  1.029e-05 -14.34 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0007814 on 321 degrees of freedom
## (13 observations deleted due to missingness)
## Multiple R-squared:  0.3903, Adjusted R-squared:  0.3884
## F-statistic: 205.5 on 1 and 321 DF, p-value: < 2.2e-16
```

```
mod_ndmi_yrs_noch_ALP <- lm(abs(ndmi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ALP"))
mod_ndmi_yrs_noch_ATL <- lm(abs(ndmi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ATL"))
mod_ndmi_yrs_noch_CON <- lm(abs(ndmi_resurvey) ~ years_between.x,
```

```

data = RF_data %>%
  filter(eunis_cambio == F) %>%
  filter(biogeo == "CON"))
mod_ndmi_yrs_noch_MED <- lm(abs(ndmi_resurvey) ~ years_between.x,
  data = RF_data %>%
  filter(eunis_cambio == F) %>%
  filter(biogeo == "MED"))
mod_ndmi_yrs_noch_PAN <- lm(abs(ndmi_resurvey) ~ years_between.x,
  data = RF_data %>%
  filter(eunis_cambio == F) %>%
  filter(biogeo == "PAN"))
Anova(mod_ndmi_yrs_noch_ALP)

```

NDMI

```

## Anova Table (Type II tests)
##
## Response: abs(ndmi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.00023943  1   78.61 < 2.2e-16 ***
## Residuals      0.00128227 421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndmi_yrs_noch_ATL)
```

```

## Anova Table (Type II tests)
##
## Response: abs(ndmi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.0047715  1  463.58 < 2.2e-16 ***
## Residuals      0.0183931 1787
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndmi_yrs_noch_CON)
```

```

## Anova Table (Type II tests)
##
## Response: abs(ndmi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.004836  1 1003.7 < 2.2e-16 ***
## Residuals      0.038923 8079
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(mod_ndmi_yrs_noch_MED)
```

```

## Anova Table (Type II tests)
##
## Response: abs(ndmi_resurvey)

```

```
##               Sum Sq Df F value    Pr(>F)
## years_between.x 0.0004134   1  14.934 0.0001818 ***
## Residuals      0.0032941 119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndmi_yrs_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndmi_resurvey)
##               Sum Sq Df F value    Pr(>F)
## years_between.x 6.7344e-05   1  82.739 < 2.2e-16 ***
## Residuals      2.6127e-04 321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(mod_ndmi_yrs_noch_ALP)
```

```
##
## Call:
## lm(formula = abs(ndmi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ALP"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0023157 -0.0010121 -0.0002147  0.0005291  0.0154085
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.596e-03  1.297e-04  20.023  <2e-16 ***
## years_between.x -7.695e-05  8.679e-06  -8.866  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001745 on 421 degrees of freedom
## (18 observations deleted due to missingness)
## Multiple R-squared:  0.1573, Adjusted R-squared:  0.1553
## F-statistic: 78.61 on 1 and 421 DF, p-value: < 2.2e-16
```

```
summary(mod_ndmi_yrs_noch_ATL)
```

```
##
## Call:
## lm(formula = abs(ndmi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ATL"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0051052 -0.0019731 -0.0003422  0.0013751  0.0307637
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.0054174  0.0001423   38.08  <2e-16 ***
## years_between.x -0.0003122  0.0000145  -21.53  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003208 on 1787 degrees of freedom
## (134 observations deleted due to missingness)
## Multiple R-squared:  0.206, Adjusted R-squared:  0.2055
## F-statistic: 463.6 on 1 and 1787 DF, p-value: < 2.2e-16
```

```
summary(mod_ndmi_yrs_noch_CON)
```

```
##
## Call:
## lm(formula = abs(ndmi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "CON"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.002865 -0.001304 -0.000588  0.000769  0.032133
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.016e-03  4.484e-05   67.28  <2e-16 ***
## years_between.x -1.516e-04  4.784e-06  -31.68  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002195 on 8079 degrees of freedom
## (351 observations deleted due to missingness)
## Multiple R-squared:  0.1105, Adjusted R-squared:  0.1104
## F-statistic: 1004 on 1 and 8079 DF, p-value: < 2.2e-16
```

```
summary(mod_ndmi_yrs_noch_MED)
```

```
##
## Call:
## lm(formula = abs(ndmi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "MED"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.006621 -0.004588 -0.001491  0.003319  0.013637
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.0075894  0.0006493   11.688 < 2e-16 ***
## years_between.x -0.0009687  0.0002507   -3.864 0.000182 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.005261 on 119 degrees of freedom
```

```
## (1 observation deleted due to missingness)
## Multiple R-squared: 0.1115, Adjusted R-squared: 0.104
## F-statistic: 14.93 on 1 and 119 DF, p-value: 0.0001818
```

```
summary(mod_ndmi_yrs_noch_PAN)
```

```
##
## Call:
## lm(formula = abs(ndmi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "PAN"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0013287 -0.0002734 -0.0001945 -0.0001082  0.0057626
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.004e-03  1.603e-04  12.498  <2e-16 ***
## years_between.x -1.080e-04  1.188e-05  -9.096  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0009022 on 321 degrees of freedom
## (13 observations deleted due to missingness)
## Multiple R-squared: 0.2049, Adjusted R-squared: 0.2025
## F-statistic: 82.74 on 1 and 321 DF, p-value: < 2.2e-16
```

```
mod_ndwi_yrs_noch_ALP <- lm(abs(ndwi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ALP"))
mod_ndwi_yrs_noch_ATL <- lm(abs(ndwi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "ATL"))
mod_ndwi_yrs_noch_CON <- lm(abs(ndwi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "CON"))
mod_ndwi_yrs_noch_MED <- lm(abs(ndwi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "MED"))
mod_ndwi_yrs_noch_PAN <- lm(abs(ndwi_resurvey) ~ years_between.x,
                             data = RF_data %>%
                               filter(eunis_cambio == F) %>%
                               filter(biogeo == "PAN"))
Anova(mod_ndwi_yrs_noch_ALP)
```

NDWI

```
## Anova Table (Type II tests)
##
## Response: abs(ndwi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.00070868    1  102.76 < 2.2e-16 ***
## Residuals      0.00290330 421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_yrs_noch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndwi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.0031087    1  431.63 < 2.2e-16 ***
## Residuals      0.0128706 1787
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_yrs_noch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndwi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.005482    1  775.02 < 2.2e-16 ***
## Residuals      0.057144 8079
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_yrs_noch_MED)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndwi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 0.0003138    1   8.0093 0.005464 **
## Residuals      0.0046627 119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod_ndwi_yrs_noch_PAN)
```

```
## Anova Table (Type II tests)
##
## Response: abs(ndwi_resurvey)
##           Sum Sq Df F value    Pr(>F)
## years_between.x 5.7612e-05    1  153.67 < 2.2e-16 ***
## Residuals      1.2035e-04 321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(mod_ndwi_yrs_noch_ALP)
```

```
##
## Call:
## lm(formula = abs(ndwi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ALP"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0039895 -0.0013711 -0.0003595  0.0010195  0.0234959
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.487e-03  1.951e-04   23.00  <2e-16 ***
## years_between.x -1.324e-04  1.306e-05  -10.14  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002626 on 421 degrees of freedom
## (18 observations deleted due to missingness)
## Multiple R-squared:  0.1962, Adjusted R-squared:  0.1943
## F-statistic: 102.8 on 1 and 421 DF,  p-value: < 2.2e-16
```

```
summary(mod_ndwi_yrs_noch_ATL)
```

```
##
## Call:
## lm(formula = abs(ndwi_resurvey) ~ years_between.x, data = RF_data %>%
##   filter(eunis_cambio == F) %>% filter(biogeo == "ATL"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0044195 -0.0016671 -0.0001417  0.0009873  0.0283831
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.671e-03  1.190e-04   39.26  <2e-16 ***
## years_between.x -2.520e-04  1.213e-05  -20.78  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002684 on 1787 degrees of freedom
## (134 observations deleted due to missingness)
## Multiple R-squared:  0.1945, Adjusted R-squared:  0.1941
## F-statistic: 431.6 on 1 and 1787 DF,  p-value: < 2.2e-16
```

```
summary(mod_ndwi_yrs_noch_CON)
```

```
##
## Call:
## lm(formula = abs(ndwi_resurvey) ~ years_between.x, data = RF_data %>%
```

```
##      filter(eunis_cambio == F) %>% filter(biogeo == "CON"))
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.003813 -0.001549 -0.000178  0.001025  0.063138
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.974e-03  5.432e-05   73.15  <2e-16 ***
## years_between.x -1.614e-04  5.797e-06  -27.84  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00266 on 8079 degrees of freedom
## (351 observations deleted due to missingness)
## Multiple R-squared:  0.08753,    Adjusted R-squared:  0.08742
## F-statistic: 775 on 1 and 8079 DF,  p-value: < 2.2e-16
```

```
summary(mod_ndwi_yrs_noch_MED)
```

```
##
## Call:
## lm(formula = abs(ndwi_resurvey) ~ years_between.x, data = RF_data %>%
##      filter(eunis_cambio == F) %>% filter(biogeo == "MED"))
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.007389 -0.004695 -0.001649  0.001435  0.013649
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.0082325  0.0007725   10.66  < 2e-16 ***
## years_between.x -0.0008440  0.0002982   -2.83  0.00546 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00626 on 119 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.06306,    Adjusted R-squared:  0.05519
## F-statistic: 8.009 on 1 and 119 DF,  p-value: 0.005464
```

```
summary(mod_ndwi_yrs_noch_PAN)
```

```
##
## Call:
## lm(formula = abs(ndwi_resurvey) ~ years_between.x, data = RF_data %>%
##      filter(eunis_cambio == F) %>% filter(biogeo == "PAN"))
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -9.616e-04 -3.059e-04 -2.394e-04  7.008e-05  2.547e-03
##
## Coefficients:
```

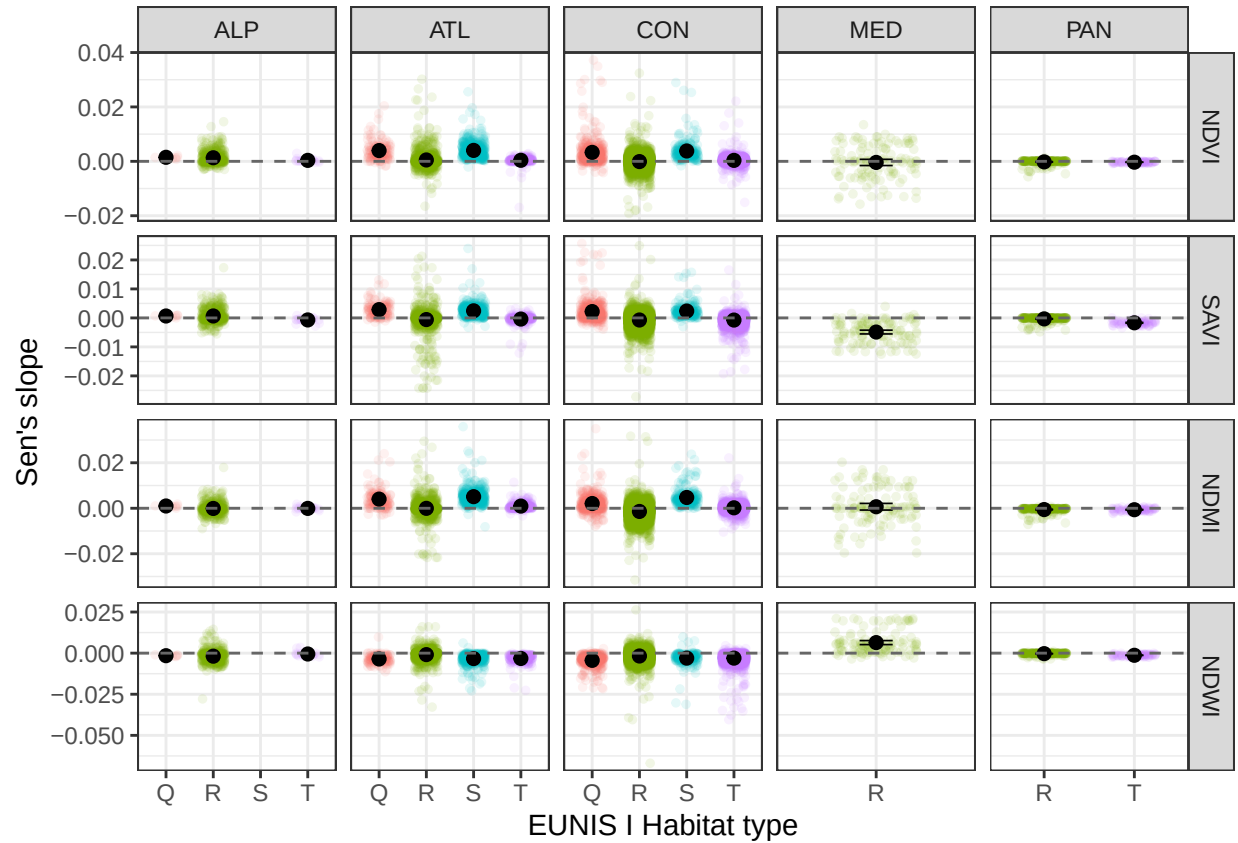
```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.861e-03  1.088e-04   17.1   <2e-16 ***
## years_between.x -9.992e-05  8.060e-06  -12.4   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0006123 on 321 degrees of freedom
## (13 observations deleted due to missingness)
## Multiple R-squared:  0.3237, Adjusted R-squared:  0.3216
## F-statistic: 153.7 on 1 and 321 DF,  p-value: < 2.2e-16
```

Appendix 4

```
app4 <- ggplot(RF_data %>% filter(eunis_cambio == F) %>%
  pivot_longer(cols = c(ndvi_resurvey, savi_resurvey,
    ndmi_resurvey, ndwi_resurvey),
    names_to = "index", values_to = "value") %>%
  mutate(index = factor(index,
    levels = c("ndvi_resurvey",
      "savi_resurvey",
      "ndmi_resurvey",
      "ndwi_resurvey"),
    labels = c("NDVI", "SAVI", "NDMI",
      "NDWI"))),
  aes(x = eunis1, y = value, fill = eunis1)) +
  geom_point(aes(color = eunis1),
    position = position_jitter(width = 0.25), size = 1, alpha = 0.1) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
    linewidth = 0.3, color = "black") +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 2,
    color = "black", fill = "black") +
  facet_grid(rows = vars(index), cols = vars(biogeo), scales = "free") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  labs(x = "EUNIS I Habitat type", y = "Sen's slope") +
  theme_bw() +
  theme(legend.position = "none")
app4
```

```
## Warning: Removed 2068 rows containing non-finite outside the scale range
## ('stat_summary()').
## Removed 2068 rows containing non-finite outside the scale range
## ('stat_summary()').
```

```
## Warning: Removed 2068 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
ggsave(app4, filename = "output/figures/app4.jpeg",
       width = 36, height = 36, units = "cm", dpi = 300)
```

```
## Warning: Removed 2068 rows containing non-finite outside the scale range
## ('stat_summary()').
```

```
## Warning: Removed 2068 rows containing non-finite outside the scale range
## ('stat_summary()').
```

```
## Warning: Removed 2068 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Plots with habitat change

```
table((RF_data %>% filter(eunis_cambio == T))$eunis_first,
      (RF_data %>% filter(eunis_cambio == T))$biogeo)
```

```
##
##      ALP ATL CON PAN
## Q      7  28  61   0
## R      9  20  73   1
## S      0  24  29   0
## T      4   0   9   0
```

Change in functional metrics by habitat type

```
mod_ndvi_eunis_first_ch_ALP <- lm(ndvi_resurvey ~ eunis_first,
                                   data = RF_data %>%
                                   filter(eunis_cambio == T) %>%
                                   filter(biogeo == "ALP"))
mod_ndvi_eunis_first_ch_ATL <- lm(ndvi_resurvey ~ eunis_first,
                                   data = RF_data %>%
                                   filter(eunis_cambio == T) %>%
                                   filter(biogeo == "ATL"))
mod_ndvi_eunis_first_ch_CON <- lm(ndvi_resurvey ~ eunis_first,
                                   data = RF_data %>%
                                   filter(eunis_cambio == T) %>%
                                   filter(biogeo == "CON"))
Anova(mod_ndvi_eunis_first_ch_ALP)
```

NDVI

```
## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 8.0200e-07 2    0.062 0.9402
## Residuals    7.1172e-05 11
```

```
Anova(mod_ndvi_eunis_first_ch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 1.435e-05 2    0.9119 0.4086
## Residuals    3.778e-04 48
```

```
Anova(mod_ndvi_eunis_first_ch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: ndvi_resurvey
##           Sum Sq Df F value    Pr(>F)
## eunis_first 0.00005906    3  5.7025 0.001052 **
## Residuals    0.00046258 134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_ndvi_eunis_first_ch_CON, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first  emmean      SE df lower.CL upper.CL
```

```
## Q          0.002051 0.000287 134  0.001484  0.00262
## R          0.000736 0.000229 134  0.000284  0.00119
## S          0.002077 0.000387 134  0.001311  0.00284
## T          0.000896 0.000702 134 -0.000493  0.00228
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R      1.31e-03 0.000367 134   3.584  0.0026
## Q - S     -2.67e-05 0.000482 134  -0.055  0.9999
## Q - T      1.15e-03 0.000759 134   1.522  0.4270
## R - S     -1.34e-03 0.000450 134  -2.981  0.0177
## R - T     -1.60e-04 0.000739 134  -0.216  0.9964
## S - T      1.18e-03 0.000802 134   1.473  0.4566
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndvi_eunis_first_ch_ALP)$r.squared
```

```
## [1] 0.01114814
```

```
summary(mod_ndvi_eunis_first_ch_ATL)$r.squared
```

```
## [1] 0.03660406
```

```
summary(mod_ndvi_eunis_first_ch_CON)$r.squared
```

```
## [1] 0.1132137
```

```
mod_savi_eunis_first_ch_ALP <- lm(savi_resurvey ~ eunis_first,
                                data = RF_data %>%
                                  filter(eunis_cambio == T) %>%
                                  filter(biogeo == "ALP"))
mod_savi_eunis_first_ch_ATL <- lm(savi_resurvey ~ eunis_first,
                                data = RF_data %>%
                                  filter(eunis_cambio == T) %>%
                                  filter(biogeo == "ATL"))
mod_savi_eunis_first_ch_CON <- lm(savi_resurvey ~ eunis_first,
                                data = RF_data %>%
                                  filter(eunis_cambio == T) %>%
                                  filter(biogeo == "CON"))
Anova(mod_savi_eunis_first_ch_ALP)
```

SAVI

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 1.1940e-06 2  0.1012 0.9046
## Residuals    6.4878e-05 11
```

```
Anova(mod_savi_eunis_first_ch_ATL)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 8.2080e-06 2  1.0457 0.3593
## Residuals    1.8837e-04 48
```

```
Anova(mod_savi_eunis_first_ch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: savi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 0.000111 3  10.244 4.06e-06 ***
## Residuals    0.000484 134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_savi_eunis_first_ch_CON, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first      emmean      SE df lower.CL upper.CL
## Q              1.53e-03 0.000293 134  0.000954  0.002114
## R              -2.88e-05 0.000234 134 -0.000492  0.000434
## S              1.18e-03 0.000396 134  0.000395  0.001963
## T              -1.81e-03 0.000718 134 -0.003234 -0.000393
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R      0.001563 0.000375 134  4.167 0.0003
## Q - S      0.000355 0.000493 134  0.720 0.8889
## Q - T      0.003348 0.000776 134  4.315 0.0002
## R - S     -0.001208 0.000460 134 -2.625 0.0471
## R - T      0.001785 0.000755 134  2.362 0.0895
## S - T      0.002993 0.000820 134  3.648 0.0021
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_savi_eunis_first_ch_ALP)$r.squared
```

```
## [1] 0.01806792
```

```
summary(mod_savi_eunis_first_ch_ATL)$r.squared
```

```
## [1] 0.04175347
```

```
summary(mod_savi_eunis_first_ch_CON)$r.squared
```

```
## [1] 0.1865566
```

```
mod_ndmi_eunis_first_ch_ALP <- lm(ndmi_resurvey ~ eunis_first,  
                                data = RF_data %>%  
                                filter(eunis_cambio == T) %>%  
                                filter(biogeo == "ALP"))  
mod_ndmi_eunis_first_ch_ATL <- lm(ndmi_resurvey ~ eunis_first,  
                                data = RF_data %>%  
                                filter(eunis_cambio == T) %>%  
                                filter(biogeo == "ATL"))  
mod_ndmi_eunis_first_ch_CON <- lm(ndmi_resurvey ~ eunis_first,  
                                data = RF_data %>%  
                                filter(eunis_cambio == T) %>%  
                                filter(biogeo == "CON"))  
Anova(mod_ndmi_eunis_first_ch_ALP)
```

NDMI

```
## Anova Table (Type II tests)  
##  
## Response: ndmi_resurvey  
##           Sum Sq Df F value Pr(>F)  
## eunis_first 7.3309e-06 2    2.02  0.179  
## Residuals   1.9960e-05 11
```

```
Anova(mod_ndmi_eunis_first_ch_ATL)
```

```
## Anova Table (Type II tests)  
##  
## Response: ndmi_resurvey  
##           Sum Sq Df F value Pr(>F)  
## eunis_first 0.00000329 2  0.2309 0.7947  
## Residuals   0.00034205 48
```

```
Anova(mod_ndmi_eunis_first_ch_CON)
```

```
## Anova Table (Type II tests)
##
## Response: ndmi_resurvey
##           Sum Sq Df F value    Pr(>F)
## eunis_first 3.496e-05  3  2.7266 0.04665 *
## Residuals   5.727e-04 134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_ndmi_eunis_first_ch_CON, pairwise ~ eunis_first)
```

```
## $emmeans
## eunis_first  emmean      SE df lower.CL upper.CL
## Q           0.001203 0.000319 134  0.000572  0.00183
## R           0.000675 0.000254 134  0.000172  0.00118
## S           0.002077 0.000431 134  0.001225  0.00293
## T           0.001399 0.000781 134 -0.000147  0.00294
##
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate      SE df t.ratio p.value
## Q - R      0.000528 0.000408 134   1.295  0.5680
## Q - S     -0.000874 0.000536 134  -1.630  0.3654
## Q - T     -0.000196 0.000844 134  -0.232  0.9956
## R - S     -0.001402 0.000501 134  -2.801  0.0295
## R - T     -0.000724 0.000822 134  -0.881  0.8147
## S - T      0.000678 0.000892 134   0.760  0.8721
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndmi_eunis_first_ch_ALP)$r.squared
```

```
## [1] 0.2686211
```

```
summary(mod_ndmi_eunis_first_ch_ATL)$r.squared
```

```
## [1] 0.009527332
```

```
summary(mod_ndmi_eunis_first_ch_CON)$r.squared
```

```
## [1] 0.05753132
```

```
mod_ndwi_eunis_first_ch_ALP <- lm(ndwi_resurvey ~ eunis_first,
                                data = RF_data %>%
                                  filter(eunis_cambio == T) %>%
```

```

                                filter(biogeo == "ALP"))
mod_ndwi_eunis_first_ch_ATL <- lm(ndwi_resurvey ~ eunis_first,
                                data = RF_data %>%
                                filter(eunis_cambio == T) %>%
                                filter(biogeo == "ATL"))
mod_ndwi_eunis_first_ch_CON <- lm(ndwi_resurvey ~ eunis_first,
                                data = RF_data %>%
                                filter(eunis_cambio == T) %>%
                                filter(biogeo == "CON"))
Anova(mod_ndwi_eunis_first_ch_ALP)

```

NDWI

```

## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 5.4530e-06 2  0.3947 0.6831
## Residuals    7.5988e-05 11

```

```
Anova(mod_ndwi_eunis_first_ch_ATL)
```

```

## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq Df F value Pr(>F)
## eunis_first 1.1472e-05 2  1.7644 0.1822
## Residuals    1.5605e-04 48

```

```
Anova(mod_ndwi_eunis_first_ch_CON)
```

```

## Anova Table (Type II tests)
##
## Response: ndwi_resurvey
##           Sum Sq  Df F value    Pr(>F)
## eunis_first 0.00010026    3  5.0905 0.002281 **
## Residuals    0.00087976 134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
emmeans(mod_ndwi_eunis_first_ch_CON, pairwise ~ eunis_first)
```

```

## $emmeans
## eunis_first  emmean      SE df lower.CL upper.CL
## Q          -0.00389 0.000395 134 -0.00467 -0.00311
## R          -0.00193 0.000315 134 -0.00255 -0.00131
## S          -0.00253 0.000534 134 -0.00359 -0.00147
## T          -0.00303 0.000968 134 -0.00494 -0.00111
##
## Confidence level used: 0.95
##

```

```
## $contrasts
## contrast estimate SE df t.ratio p.value
## Q - R -0.001963 0.000506 134 -3.881 0.0009
## Q - S -0.001364 0.000665 134 -2.052 0.1746
## Q - T -0.000867 0.001050 134 -0.828 0.8409
## R - S 0.000599 0.000620 134 0.966 0.7691
## R - T 0.001096 0.001020 134 1.076 0.7046
## S - T 0.000497 0.001110 134 0.449 0.9697
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Proportion of variance explained:

```
summary(mod_ndwi_eunis_first_ch_ALP)$r.squared
```

```
## [1] 0.06695727
```

```
summary(mod_ndwi_eunis_first_ch_ATL)$r.squared
```

```
## [1] 0.06848097
```

```
summary(mod_ndwi_eunis_first_ch_CON)$r.squared
```

```
## [1] 0.102307
```

Session info

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
## LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Spain.utf8 LC_CTYPE=Spanish_Spain.utf8
## [3] LC_MONETARY=Spanish_Spain.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Spain.utf8
##
## time zone: Europe/Madrid
## tzcode source: internal
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
```

```

## [1] stringr_1.6.0    patchwork_1.3.2 ggeffects_2.3.2 multcomp_1.4-30
## [5] TH.data_1.1-5    MASS_7.3-65      survival_3.8-6  mvtnorm_1.4-1
## [9] emmeans_2.0.4    car_3.1-5        carData_3.0-6   gghalves_0.1.4
## [13] ggplot2_4.0.3    corrplot_0.95    here_1.0.2      readr_2.2.0
## [17] tidyr_1.3.2      dplyr_1.2.1
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1  farver_2.1.2      S7_0.2.2          fastmap_1.2.0
## [5] digest_0.6.39     rpart_4.1.27      estimability_2.0.0 lifecycle_1.0.5
## [9] cluster_2.1.8.2   magrittr_2.0.5    compiler_4.6.1    rlang_1.2.0
## [13] Hmisc_5.2-6       tools_4.6.1       utf8_1.2.6        yaml_2.3.12
## [17] data.table_1.18.4 knitr_1.51        labeling_0.4.3    htmlwidgets_1.6.4
## [21] bit_4.6.0         RColorBrewer_1.1-3 abind_1.4-8        withr_3.0.3
## [25] foreign_0.8-91    purrr_1.2.2       nnet_7.3-20       grid_4.6.1
## [29] xtable_1.8-8      colorspace_2.1-3  scales_1.4.0      insight_1.5.4
## [33] cli_3.6.6         rmarkdown_2.31    crayon_1.5.3      ragg_1.5.2
## [37] generics_0.1.4    otel_0.2.0        rstudioapi_0.19.0 tzdb_0.5.0
## [41] splines_4.6.1     parallel_4.6.1    base64enc_0.1-6   vctrs_0.7.3
## [45] Matrix_1.7-5      sandwich_3.1-1    hms_1.1.4         bit64_4.8.2
## [49] Formula_1.2-5     htmlTable_2.5.0   systemfonts_1.3.2 glue_1.8.1
## [53] codetools_0.2-20  stringi_1.8.7     gtable_0.3.6      tibble_3.3.1
## [57] pillar_1.11.1     htmltools_0.5.9   R6_2.6.1          textshaping_1.0.5
## [61] rprojroot_2.1.1   vroom_1.7.1       evaluate_1.0.5    lattice_0.22-9
## [65] backports_1.5.1   coda_0.19-4.1     gridExtra_2.3.1   nlme_3.1-169
## [69] checkmate_2.3.4   mgcv_1.9-4        xfun_0.59         zoo_1.8-15
## [73] pkgconfig_2.0.3

```